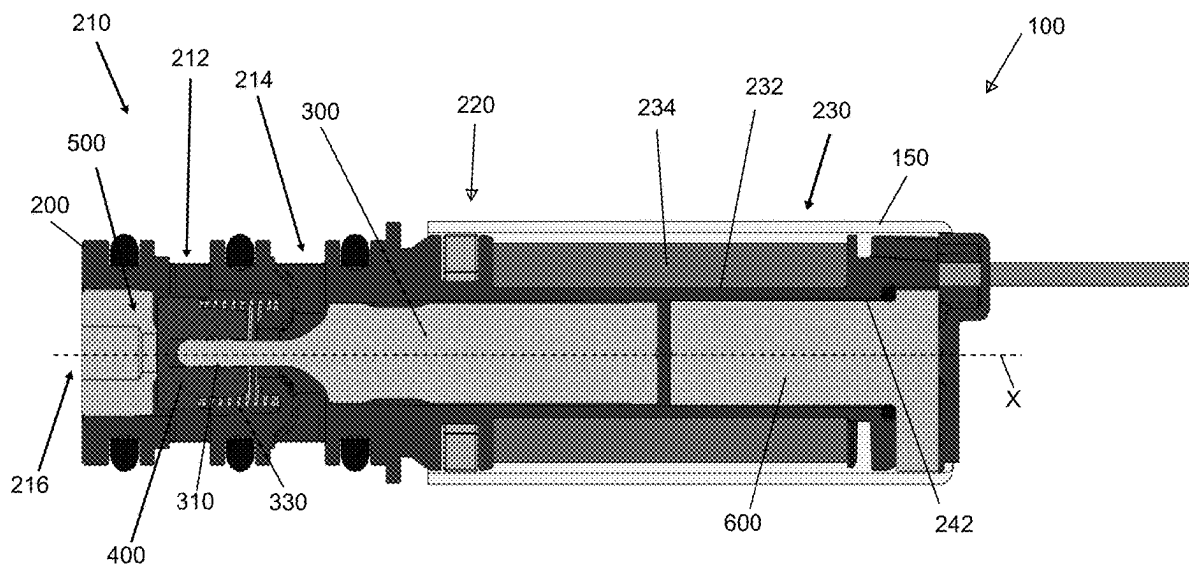


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A method of assembling a valve can include monitoring a first fluid flow between a first port and a second port of the valve and/or pressing a poppet onto a plunger of the valve until the first fluid flow stops. The method can include monitoring a second fluid flow between the first port and a third port of the valve, energizing a coil of the valve, and pressing a cap into a body of the valve until the second fluid flow stops or until a target rate of the second fluid flow is achieved.



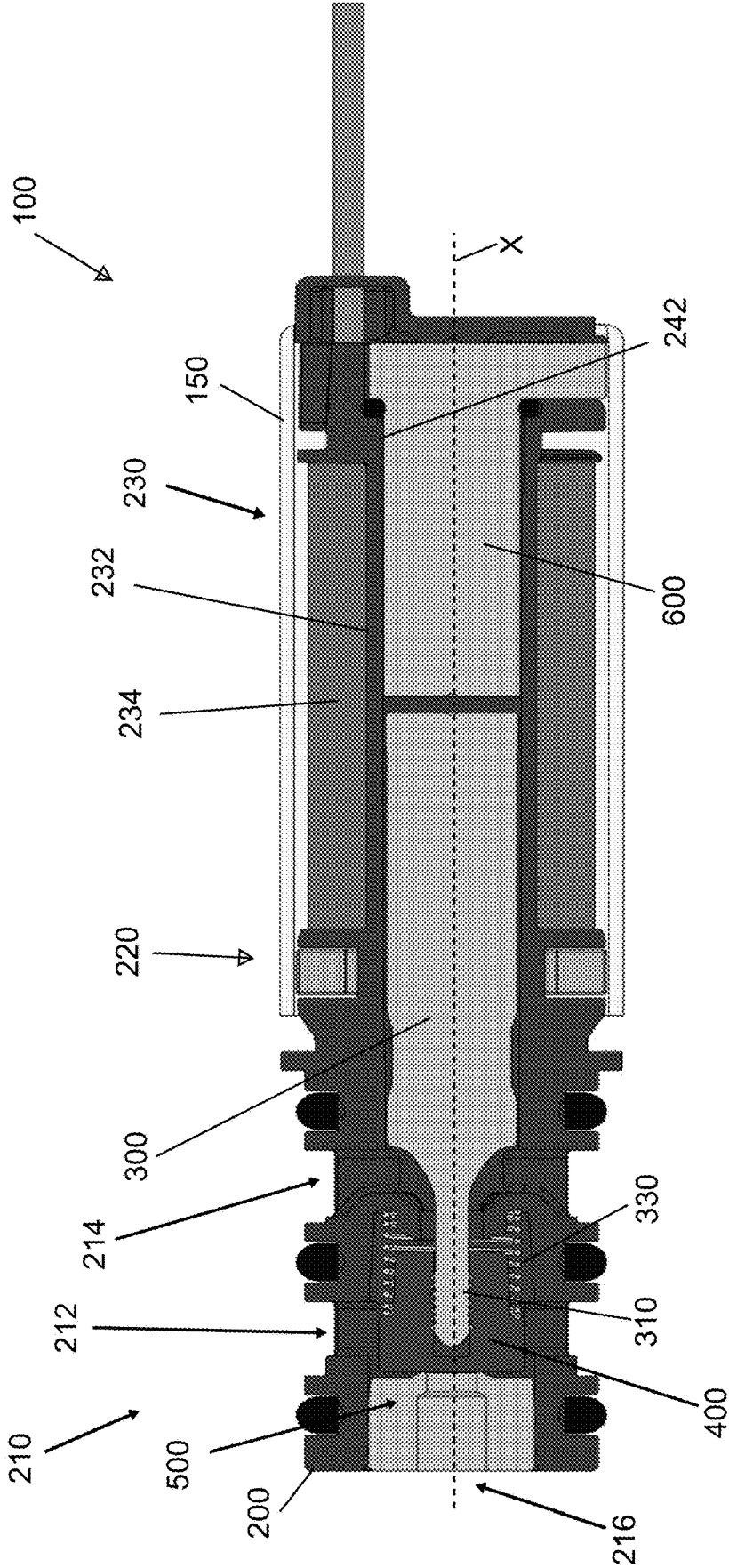


FIG. 1

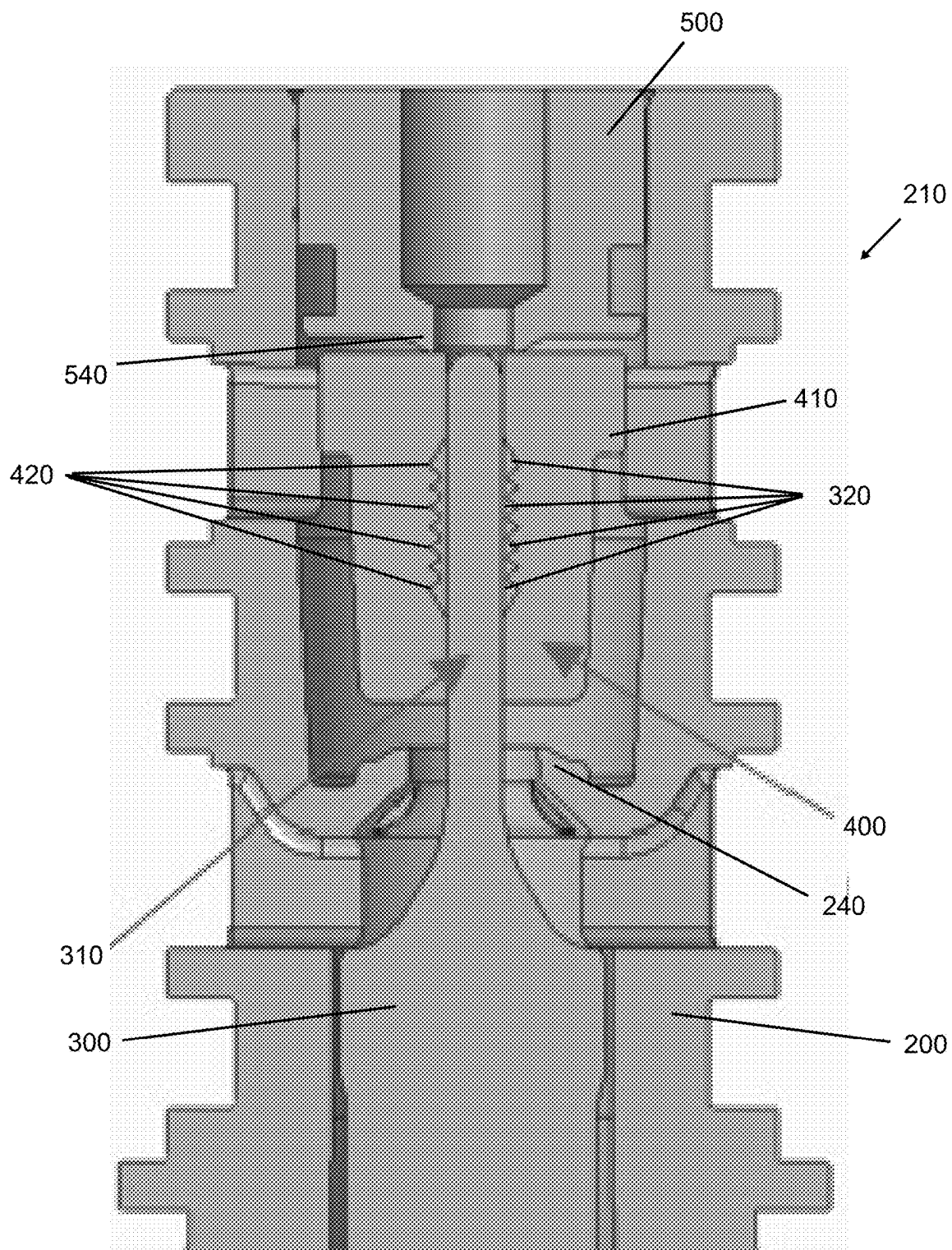
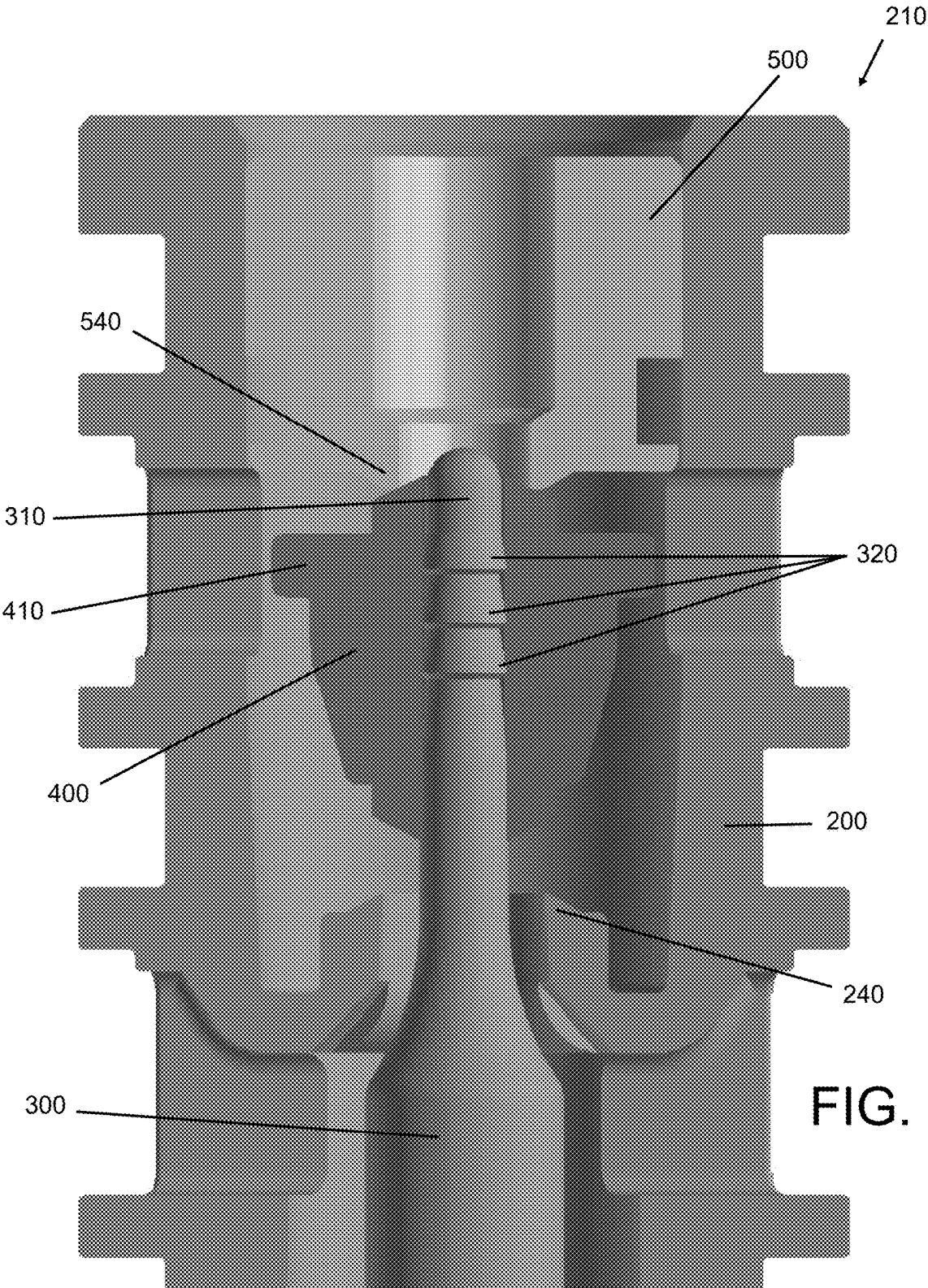


FIG. 2



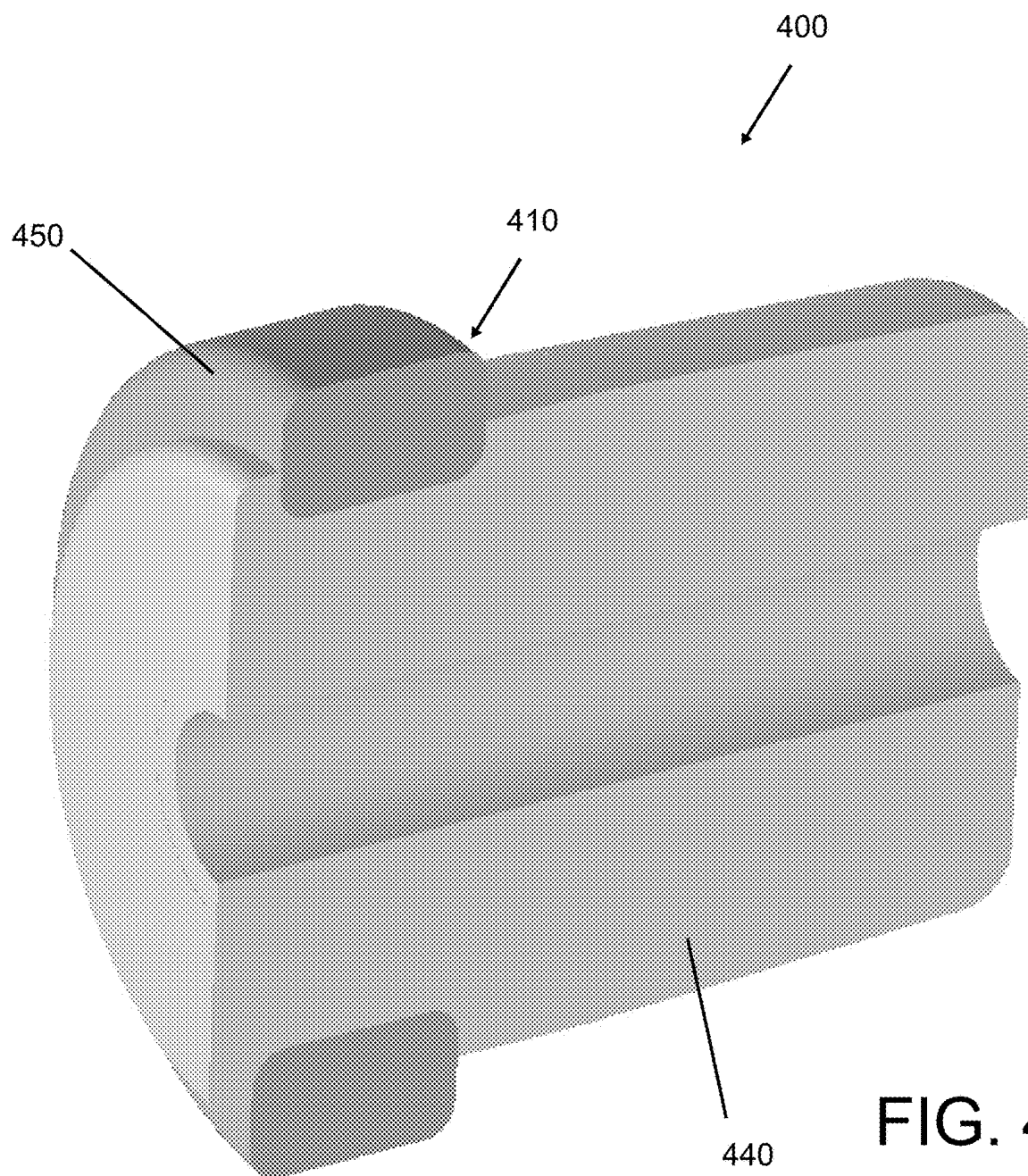
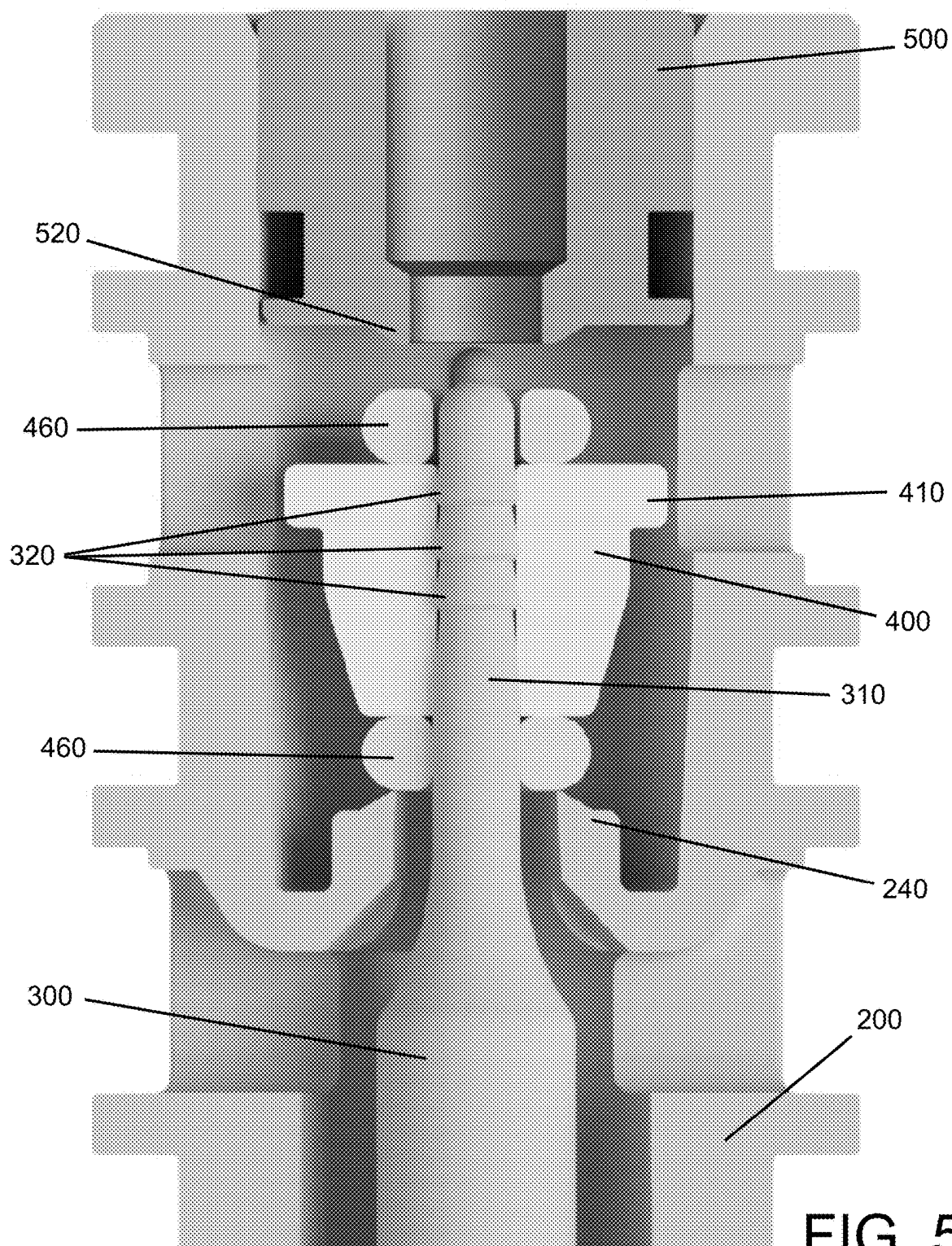
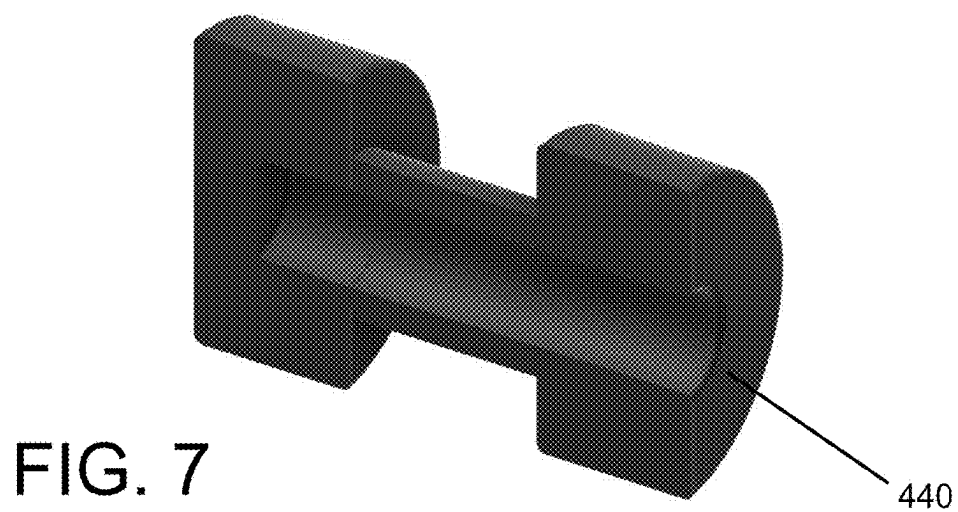
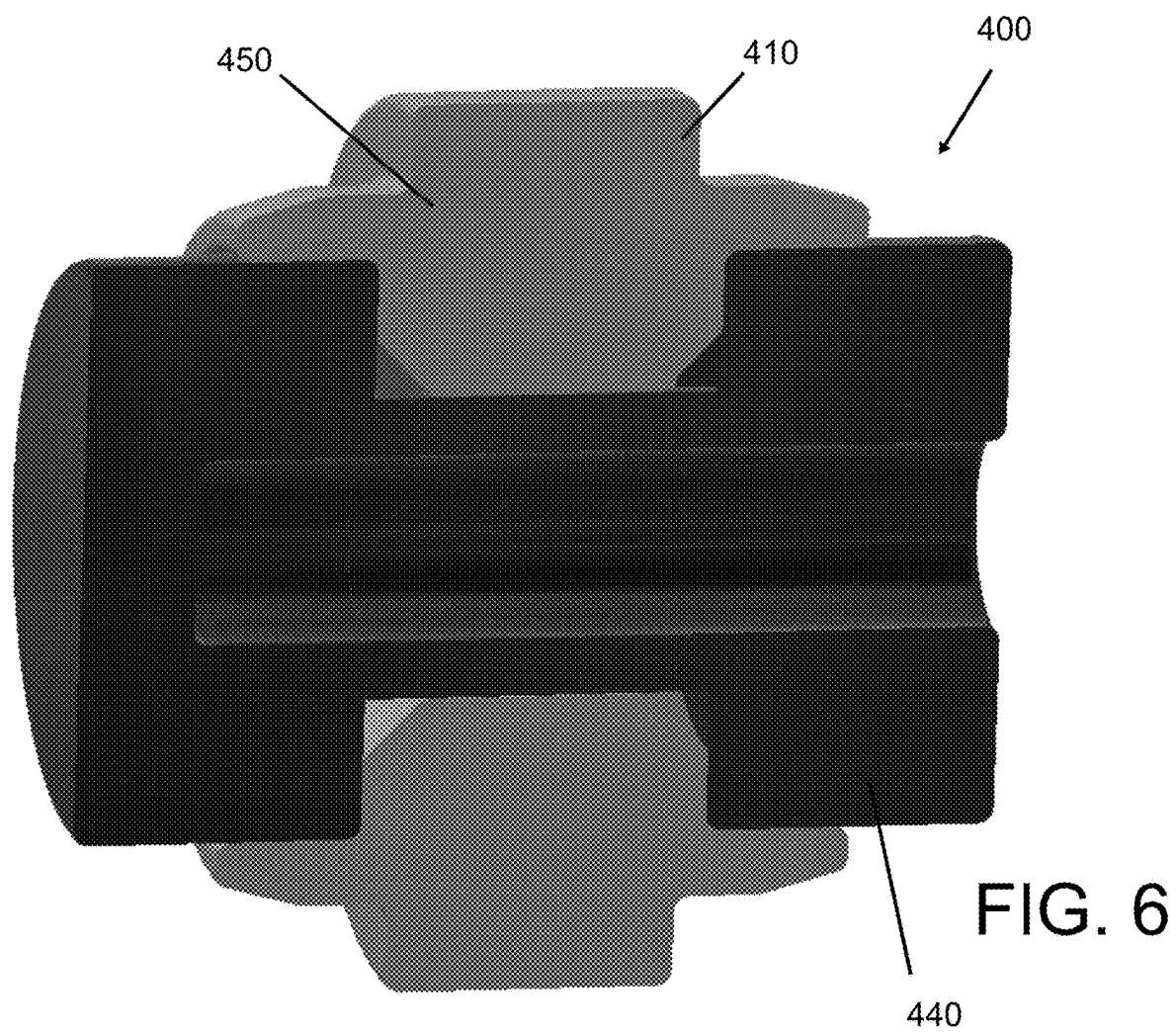


FIG. 4





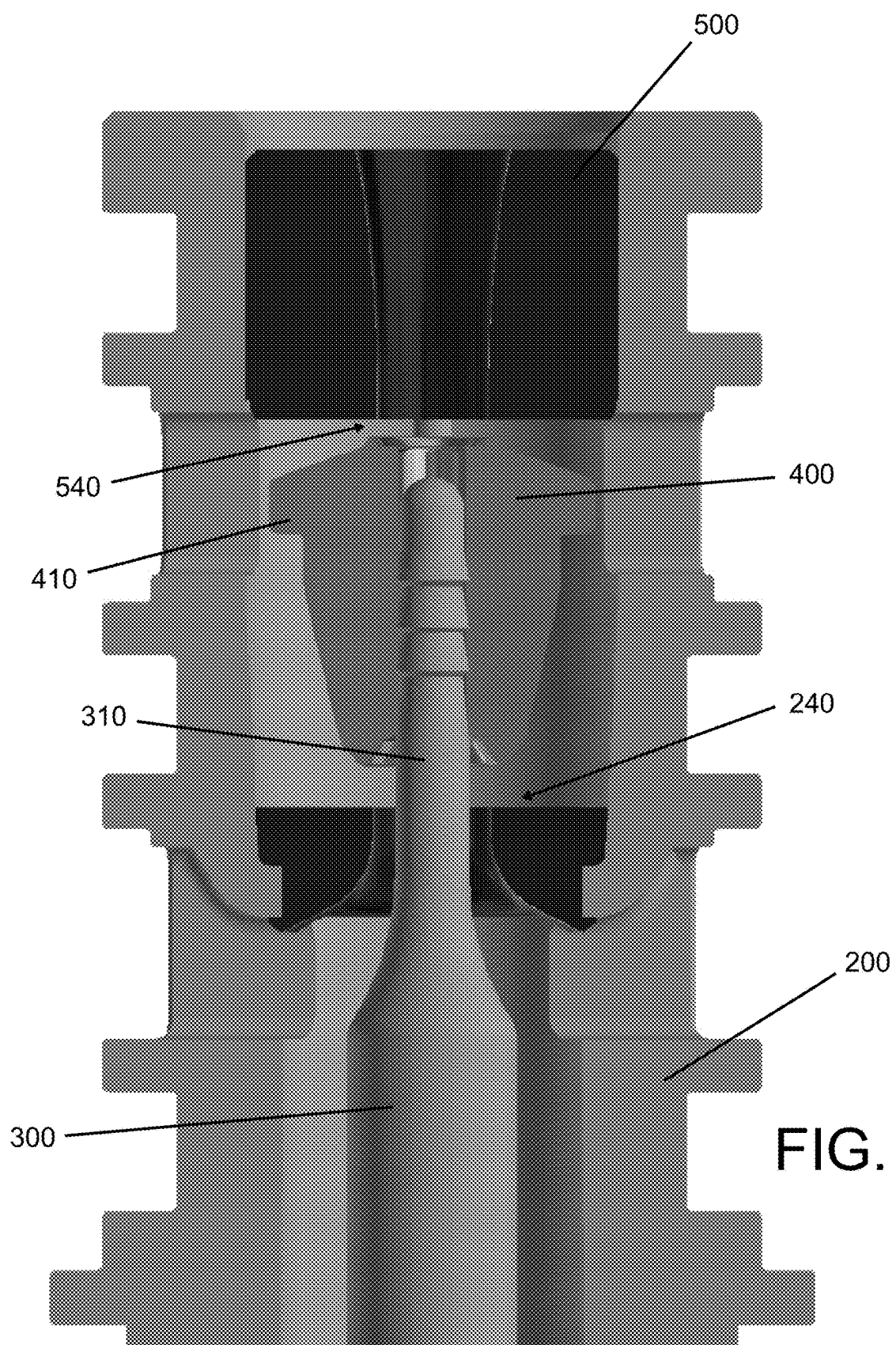


FIG. 8

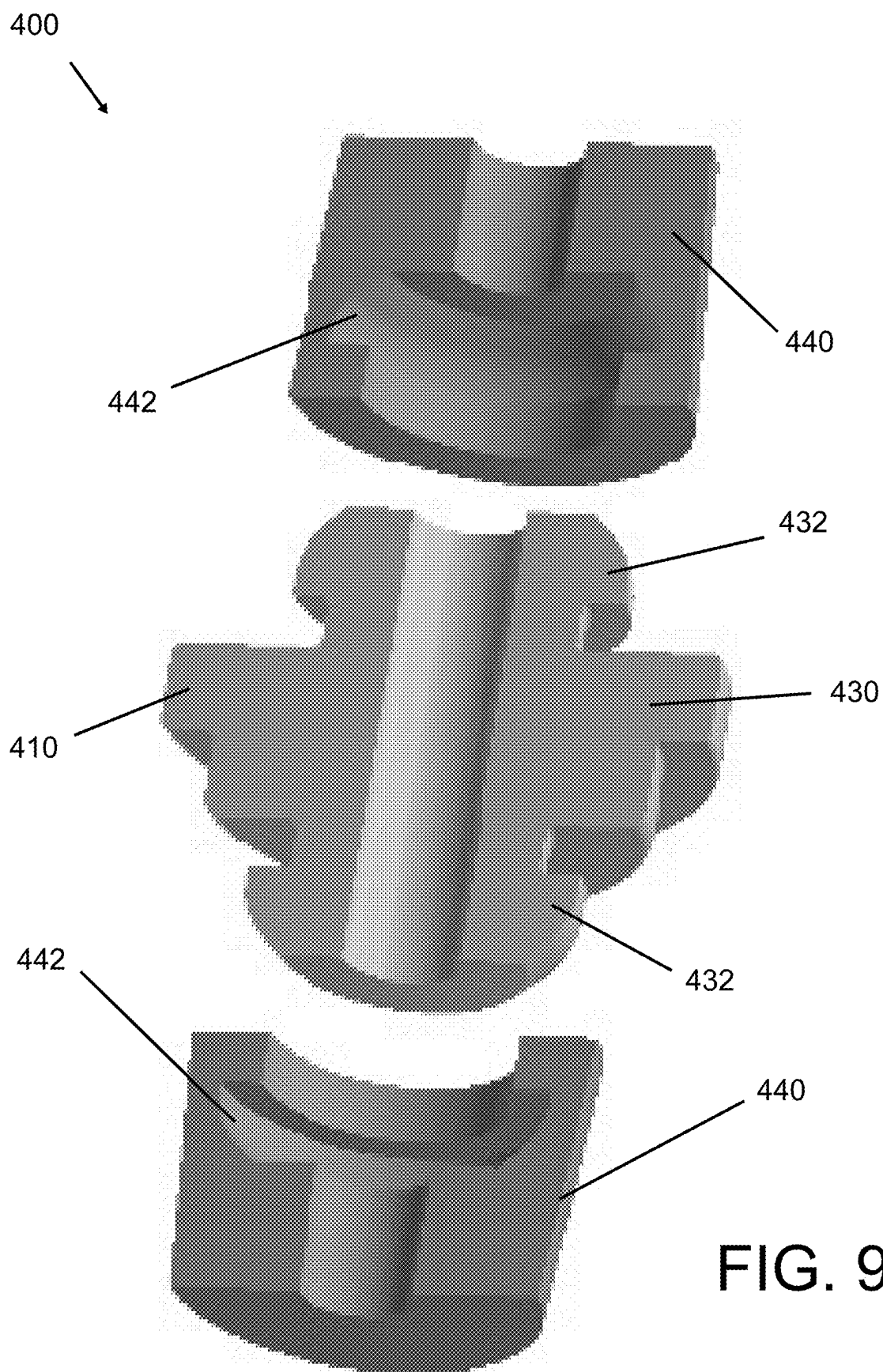
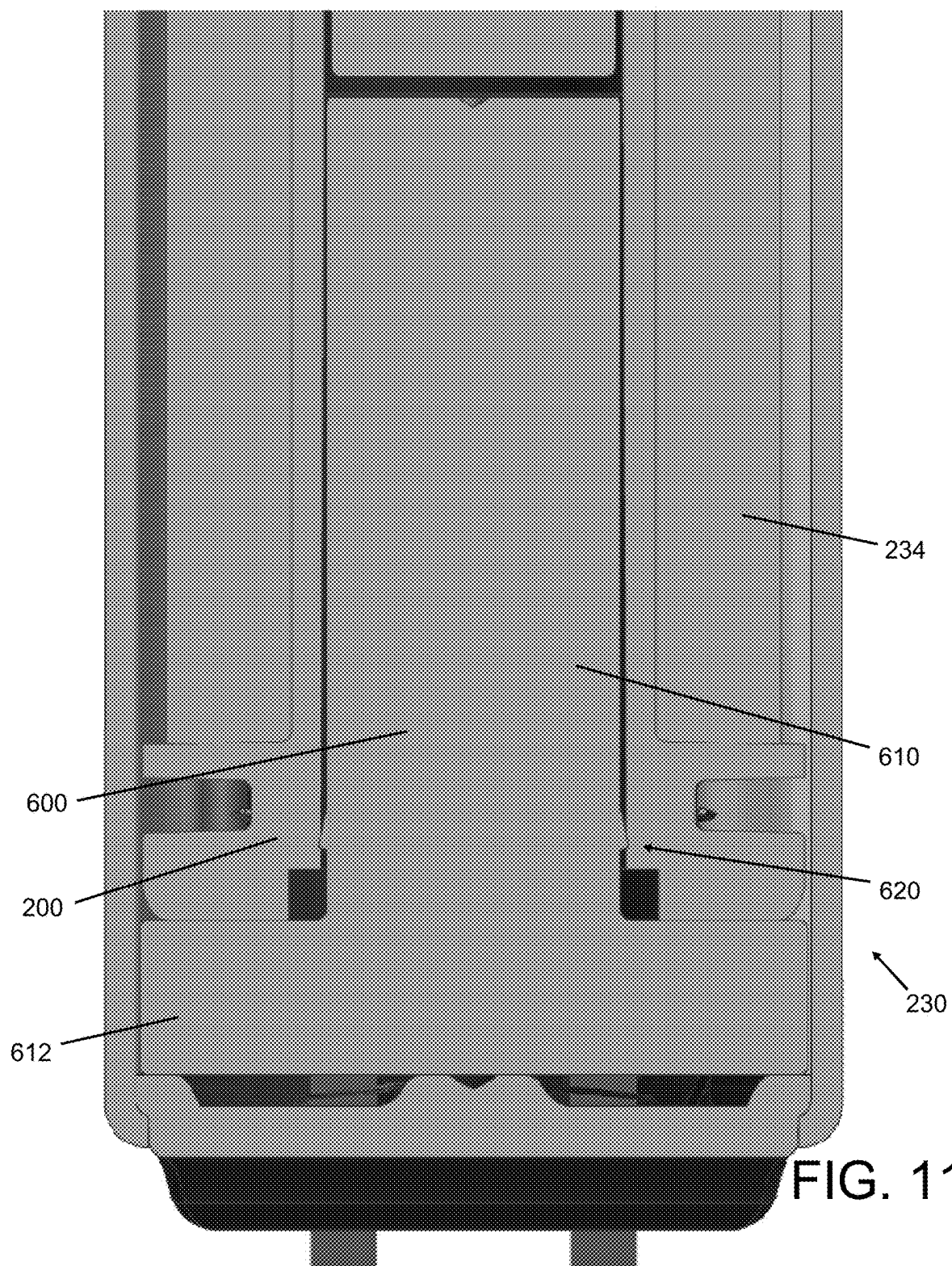
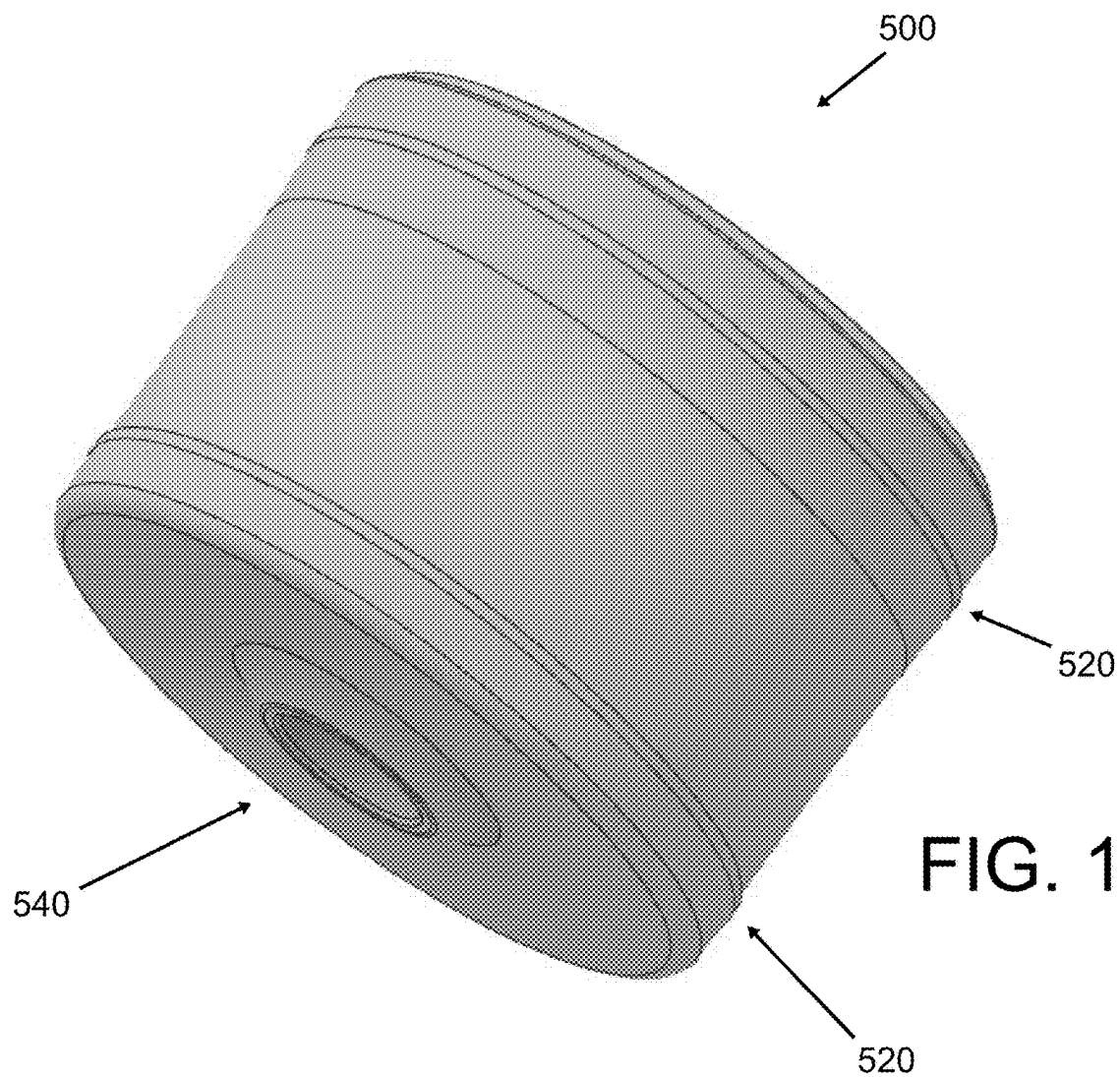


FIG. 9

FIG. 10





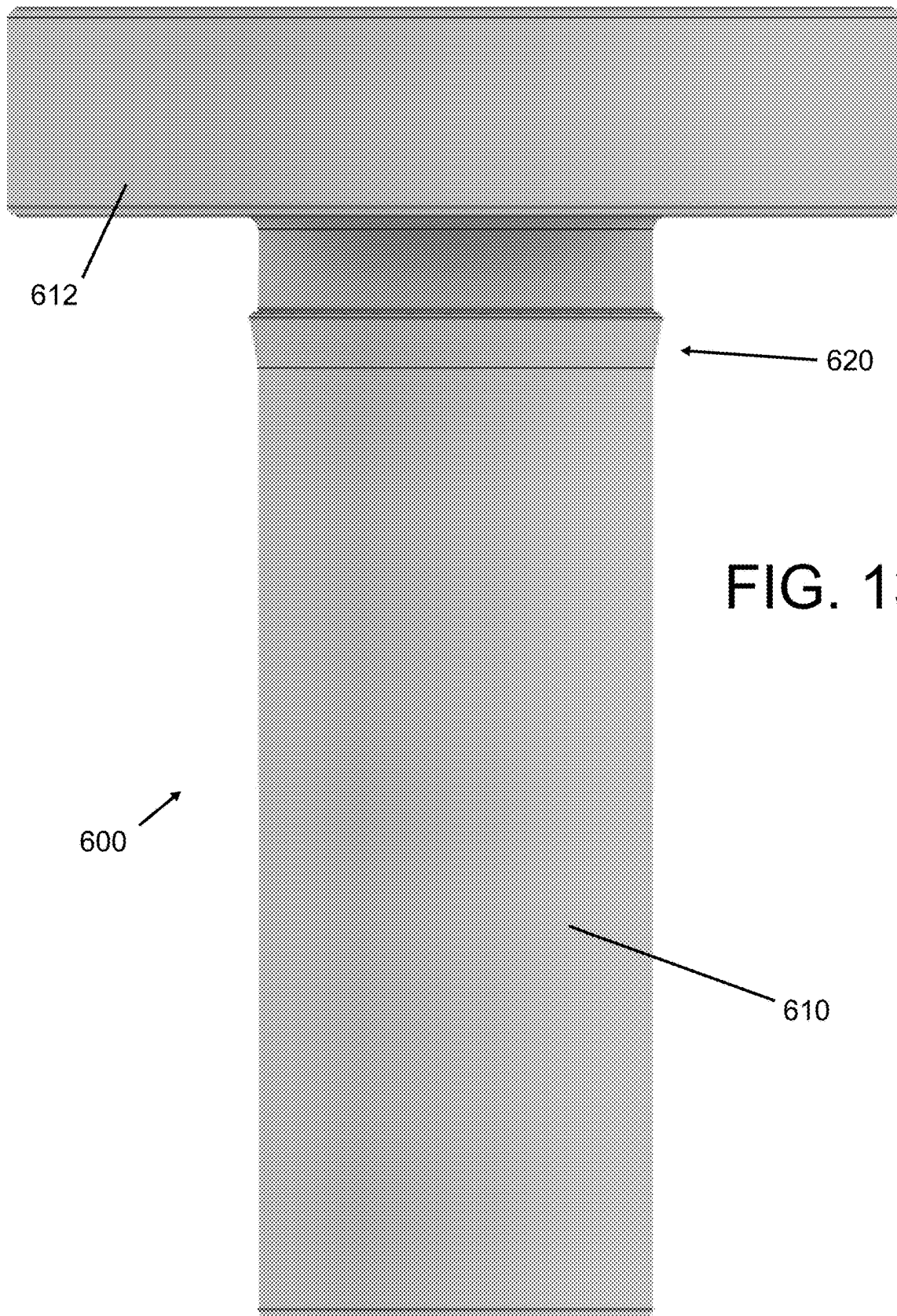
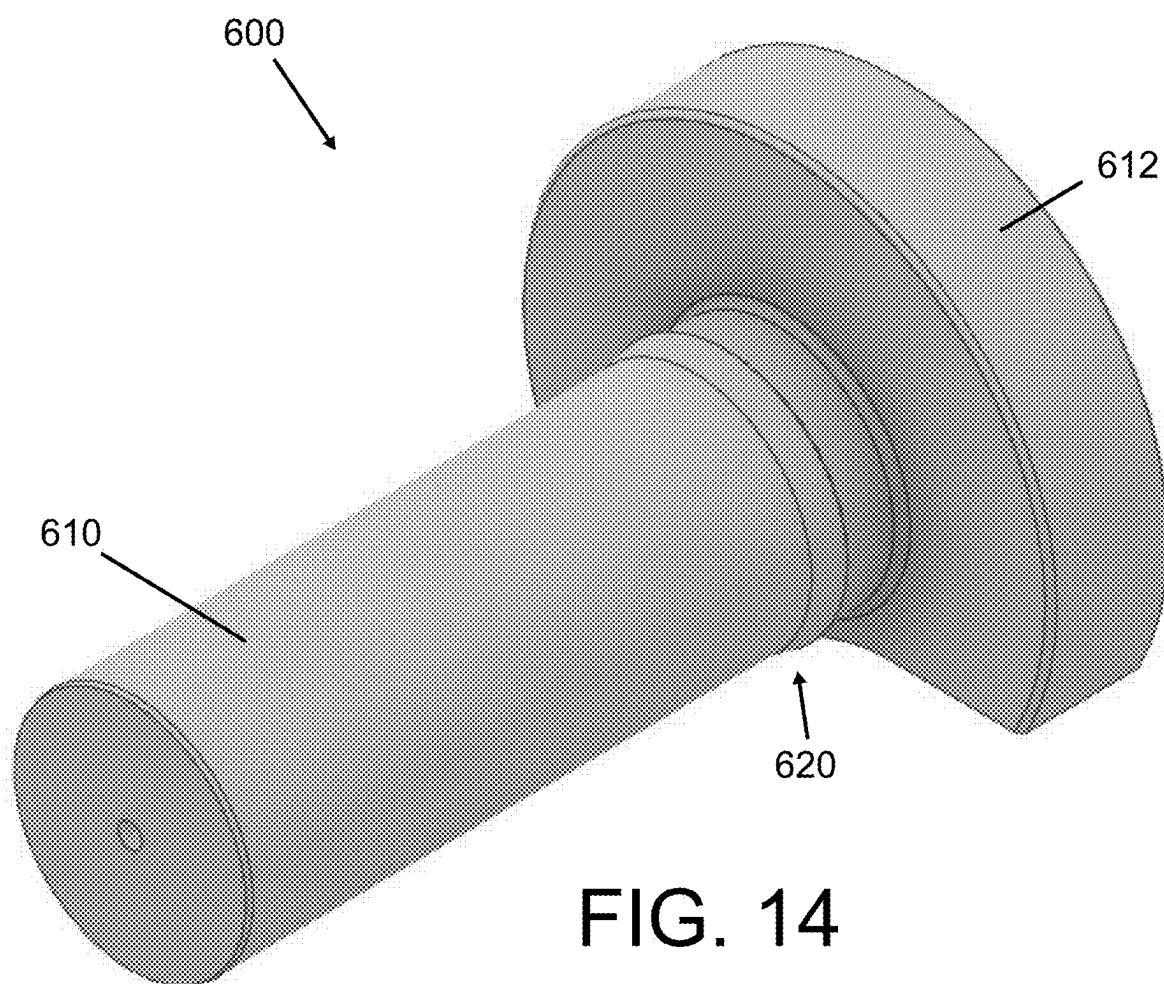
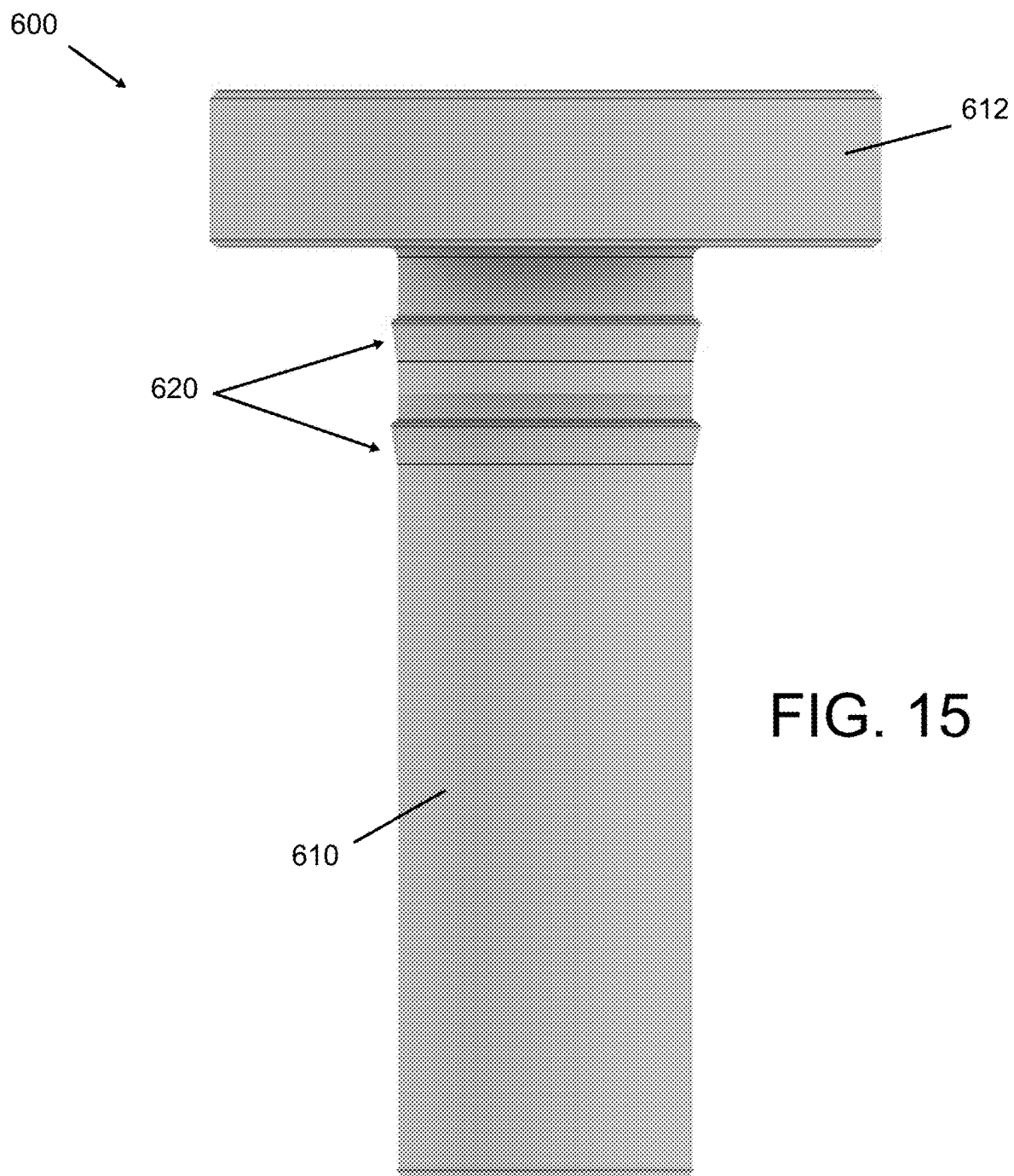
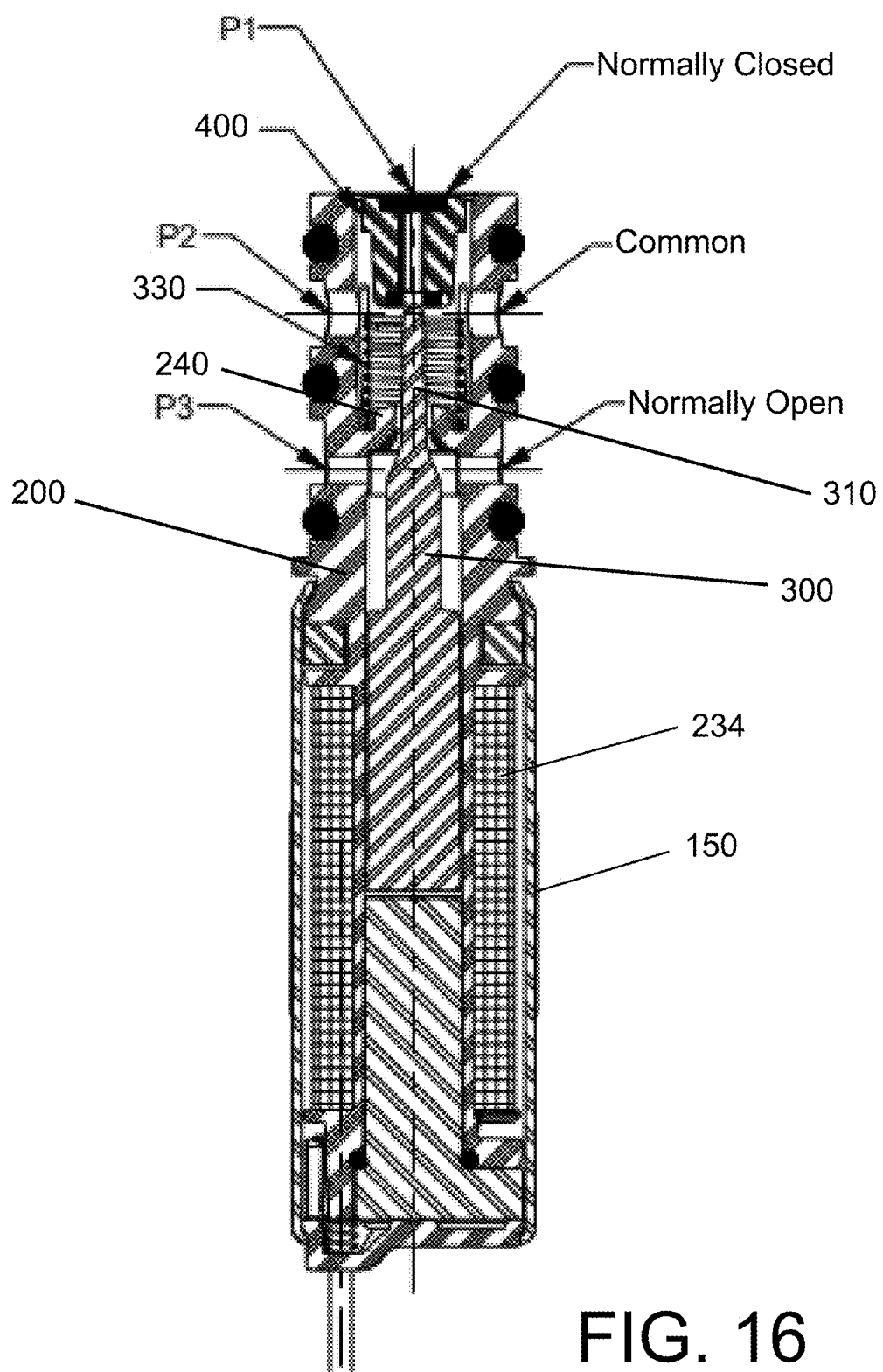


FIG. 13







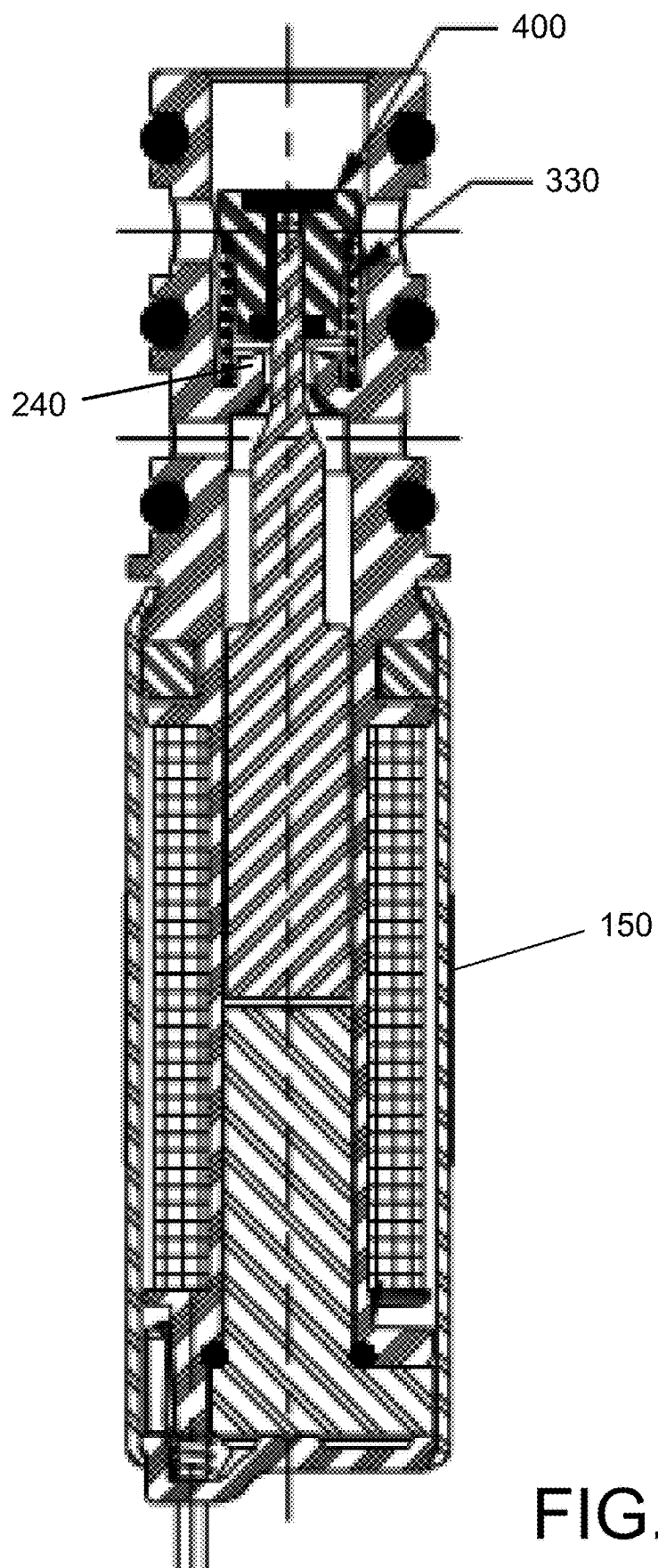
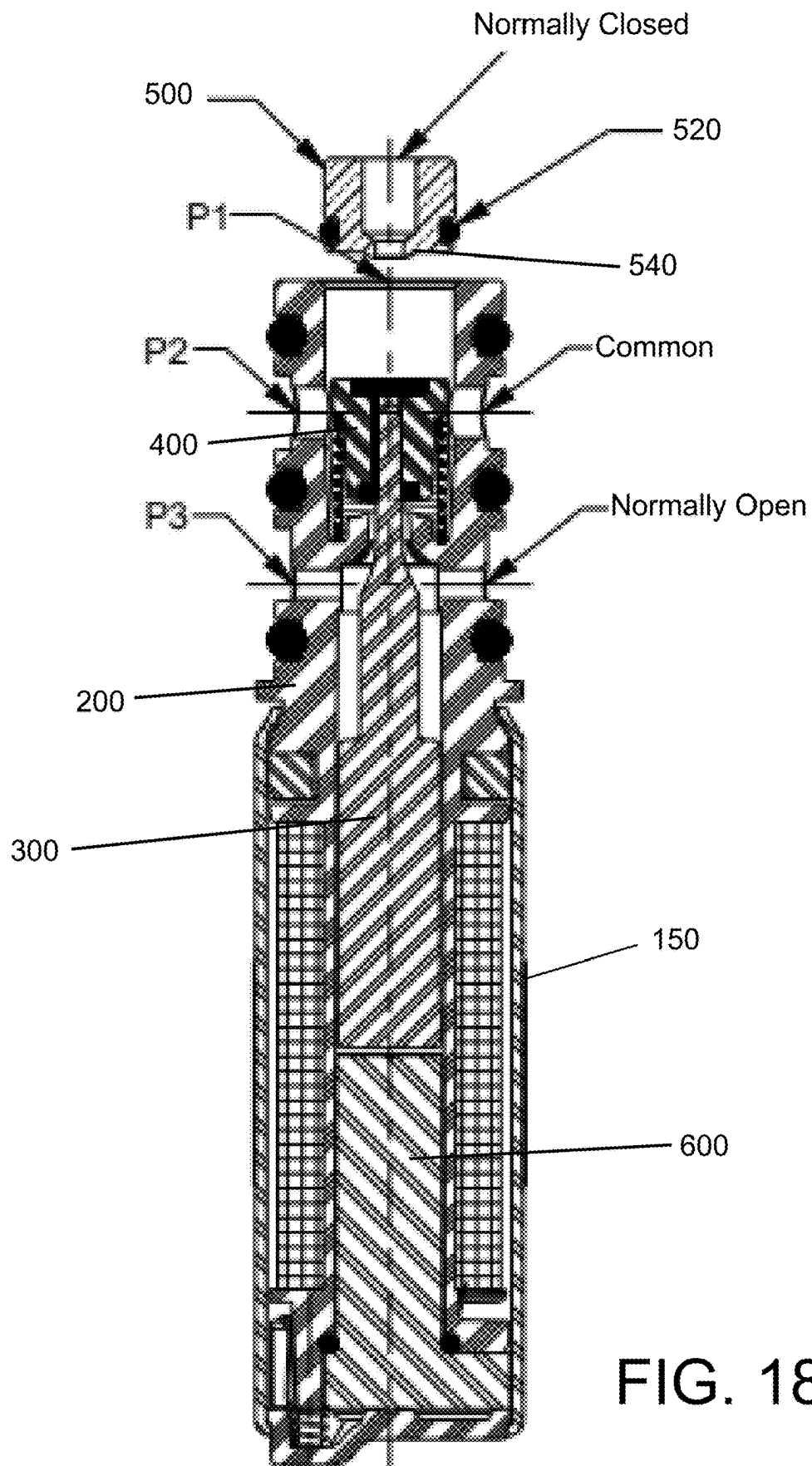
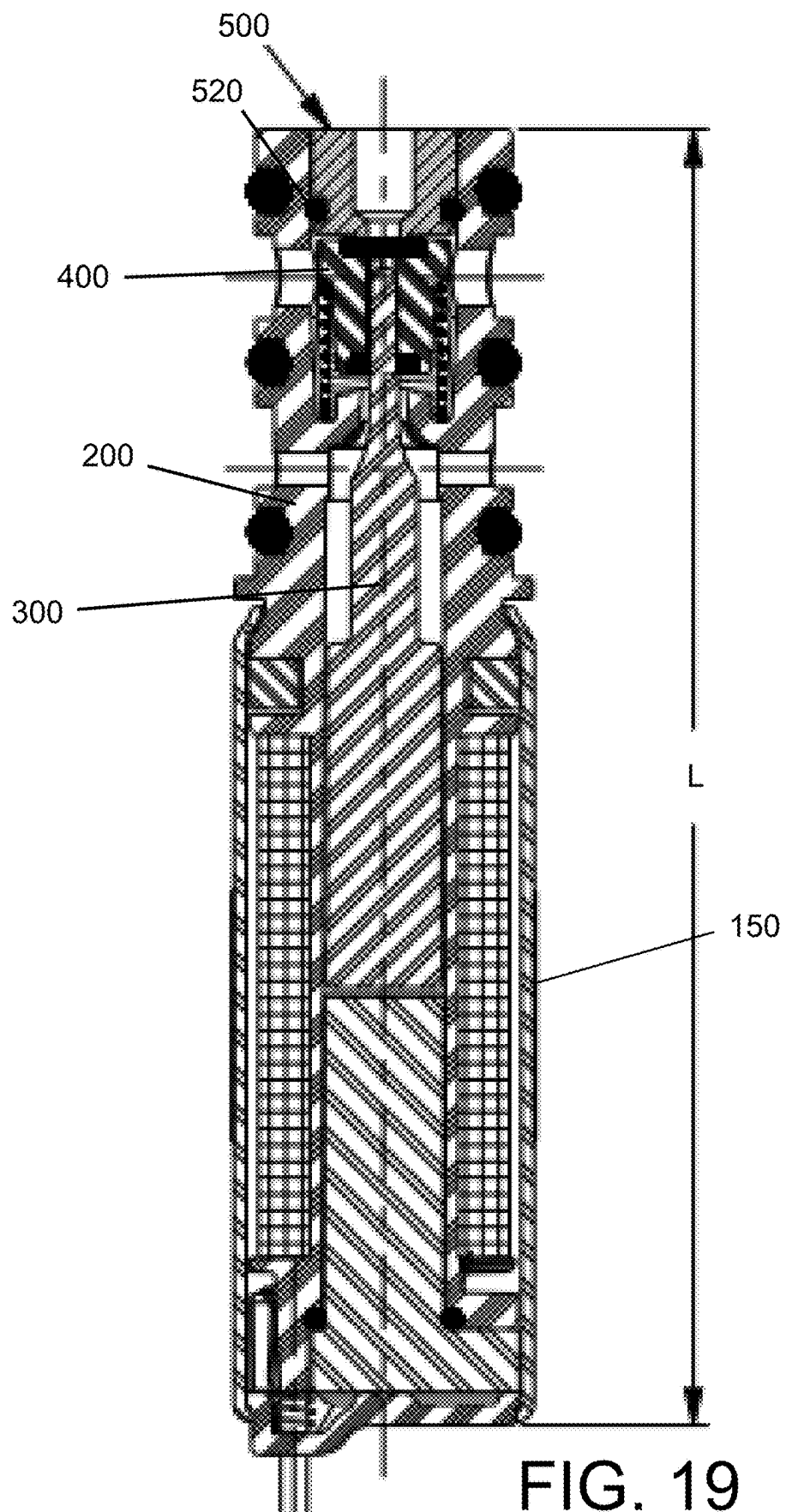


FIG. 17





700 ↗

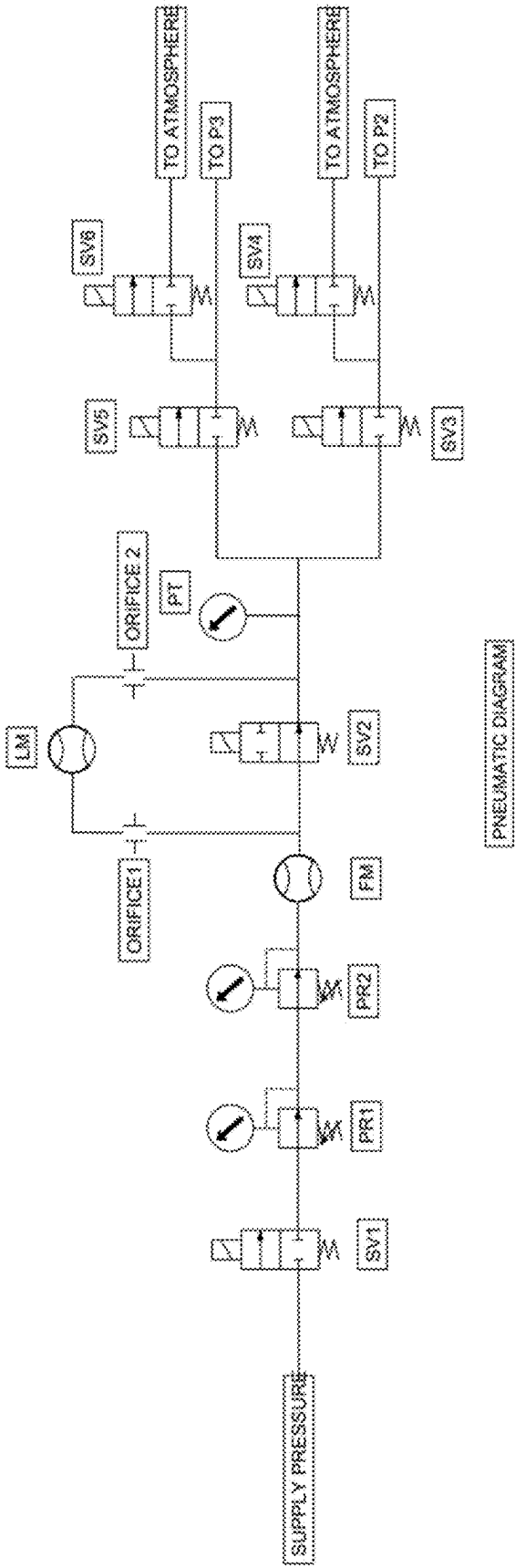


FIG. 20

PRESS TO SPECIFICATION SOLENOID VALVE ASSEMBLY

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to the commonly owned applications entitled “SOLENOID VALVE WITH BARBED SEALING ASSEMBLY,” “VALVE WITH IMPROVED MAGNETIC FLUX INTERFACE,” “SOLENOID VALVE POPPET AFFIXMENT,” and “SOLENOID PLUNGER ALIGNMENT,” respectively, each filed on the same day as the instant application, and the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates generally to solenoid valves and more specifically relates to miniature solenoid valves.

Description of the Related Art

[0003] Solenoid valves typically have several components that are each manufactured to their own tolerances. As valves are assembled those tolerances can stack up, or add up, such that even properly assembled valves can experience flow variation, variation in leakage, and variation in pull-in or drop-out current. To mitigate against these effects, component tolerances can be tightened; however, this adds cost to the manufacture of the valve.

SUMMARY OF THE INVENTION

[0004] Applicants have created new and useful devices, systems and methods for solenoid valves. In at least one embodiment, a valve can include a valve body having a port section and a coil support section extending along a longitudinal axis, a plunger within the valve body and extending at least partially through the coil support section, a poppet press-fit onto the plunger, or any combination thereof. In at least one embodiment, the body can include a first valve seat. In at least one embodiment, the plunger can include a stem extending through the first valve seat into the port section. In at least one embodiment, the poppet can selectively engage the first valve seat to open and close a first flow path through the port section of the valve body. In at least one embodiment, a stroke of the plunger can be adjusted by adjusting a position of the poppet on the plunger. In at least one embodiment, a flow rate through the first flow path is configured to be adjusted by adjusting a position of the poppet on the plunger.

[0005] In at least one embodiment, the valve can include a cap press-fit at least partially into the port section of the valve body. In at least one embodiment, the cap can include a second valve seat that is configured to be selectively engaged by the poppet to selectively open and close a second flow path through the port section of the valve body. In at least one embodiment, the stroke of the plunger can be adjusted by adjusting a second position of the cap in the body. In at least one embodiment, a flow rate through the second flow path can be adjusted by adjusting a second position of the cap in the body.

[0006] In at least one embodiment, a method of assembling a valve can include monitoring a first flow, or fluid

flow, between a first port and a second port of the valve and/or pressing a poppet onto a plunger of the valve until the first flow stops. In at least one embodiment, the method can include monitoring a second fluid flow between the first port and a third port of the valve, energizing a coil of the valve, pressing a cap into a body of the valve until the second fluid flow stops, or any combination thereof. In at least one embodiment, the method can include monitoring a second fluid flow between the first port and a third port of the valve and/or pressing a cap into a body of the valve until a target rate of the second fluid flow is achieved. In at least one embodiment, the first port is a common port, the second port is a normally closed port, and the third port is a normally open port.

[0007] In at least one embodiment, the method can include monitoring a power consumption of the coil and repeating the step of pressing the cap into the body, until the power consumption matches a setpoint. In at least one embodiment, the method can include energizing a coil of the valve, de-energizing the coil of the valve, monitoring for a sound caused by a plunger of the valve contacting a stop of the valve, or any combination thereof. In at least one embodiment, the method can include repeating, until the sound caused by the plunger contacting the stop is no longer detected, the steps of pressing the poppet onto the plunger, energizing the coil of the valve, de-energizing the coil of the valve, monitoring for the sound caused by the plunger contacting the stop, or any combination thereof.

[0008] In at least one embodiment, a method of assembling a valve can include monitoring a first flow between a common port and a normally closed port of the valve, pressing a poppet onto a plunger of the valve until the first flow stops, monitoring a second fluid flow between the common port and a normally open port of the valve, energizing a coil of the valve, pressing a cap into a body of the valve until the second fluid flow stops, or any combination thereof. In at least one embodiment, the method can include monitoring a power consumption of the coil and repeating the step of pressing the cap into the body, until the power consumption matches a setpoint. In at least one embodiment, the method can include energizing a coil of the valve, de-energizing the coil of the valve, monitoring for a sound caused by a plunger of the valve contacting a stop of the valve, or any combination thereof. In at least one embodiment, the method can include repeating, until the sound caused by the plunger contacting the stop is no longer detected, the steps of pressing the poppet onto the plunger, energizing the coil of the valve, de-energizing the coil of the valve, monitoring for the sound caused by the plunger contacting the stop, or any combination thereof. In at least one embodiment, the poppet can be pressed onto the plunger before the cap is pressed into the body.

[0009] In at least one embodiment, a method of assembling a valve can include monitoring a fluid flow between two or more ports of the valve, energizing a coil of the valve, pressing a cap into a body of the valve until the fluid flow stops, or any combination thereof. In at least one embodiment, the method can include monitoring a power consumption of the coil and repeating the step of pressing the cap into the body, until the power consumption matches a setpoint. In at least one embodiment, the method can include energizing a coil of the valve, de-energizing the coil of the valve, monitoring for a sound caused by a plunger of the valve contacting one or more other components, such as a stop of

the valve, or any combination thereof. In at least one embodiment, the method can include repeating, until a sound caused by the plunger is no longer detected (or, e.g., falls below a threshold or within an acceptable range), the steps of pressing a poppet onto a plunger of the valve, energizing the coil of the valve, de-energizing the coil of the valve, monitoring for the sound caused by the plunger, or any combination thereof. In at least one embodiment, the poppet can be pressed onto the plunger before the cap is pressed into the body.

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is a cross sectional view of one of many embodiments of a solenoid valve according to the disclosure.

[0011] FIG. 2 is a cross sectional view of a portion of one of many embodiments of a solenoid valve according to the disclosure.

[0012] FIG. 3 is a cross sectional view of a portion of another one of many embodiments of a solenoid valve according to the disclosure.

[0013] FIG. 4 is a cross sectional perspective view of one of many embodiments of a poppet according to the disclosure.

[0014] FIG. 5 is a cross sectional view of a portion of yet another one of many embodiments of a solenoid valve according to the disclosure.

[0015] FIG. 6 is a cross sectional perspective view of another one of many embodiments of a poppet according to the disclosure.

[0016] FIG. 7 is a cross sectional perspective view of a portion of the poppet of FIG. 6.

[0017] FIG. 8 is a cross sectional view of a portion of still another of many embodiments of a solenoid valve according to the disclosure.

[0018] FIG. 9 is an exploded cross sectional view of a portion of one of many embodiments of a poppet according to the disclosure.

[0019] FIG. 10 is a cross sectional view of a portion of one of many embodiments of a solenoid valve according to the disclosure.

[0020] FIG. 11 is a cross sectional view of another portion of one of many embodiments of a solenoid valve according to the disclosure.

[0021] FIG. 12 is a perspective view of one of many embodiments of a cap according to the disclosure.

[0022] FIG. 13 is an elevation view of one of many embodiments of a stop according to the disclosure.

[0023] FIG. 14 is a perspective view of the stop of FIG. 13.

[0024] FIG. 15 is an elevation view of another one of many embodiments of a stop according to the disclosure.

[0025] FIG. 16 is a cross sectional view illustrating one of many embodiments of an assembly method according to the disclosure, prior to the poppet being pressed onto the plunger.

[0026] FIG. 17 is a cross sectional view illustrating another of many embodiments of an assembly method according to the disclosure, showing the poppet pressed onto the plunger.

[0027] FIG. 18 is a cross sectional view illustrating yet another of many embodiments of an assembly method according to the disclosure, prior to the cap being pressed into the valve body.

[0028] FIG. 19 is a cross sectional view illustrating still another of many embodiments of an assembly method according to the disclosure, showing the cap pressed into the valve body.

[0029] FIG. 20 is a pneumatic diagram of one of many embodiments of an assembly system according to the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0030] The figures described above and the written description of specific structures and functions below are not presented to limit the scope of what Applicants have invented or the scope of the appended claims. Rather, the figures and written description are provided to teach any person skilled in the art to make and use the inventions for which patent protection is sought. Those skilled in the art will appreciate that not all features of a commercial embodiment of the inventions are described or shown for the sake of clarity and understanding. Persons of skill in this art will also appreciate that the development of an actual commercial embodiment incorporating aspects of the present inventions will require numerous implementation-specific decisions to achieve the developer's ultimate goal for the commercial embodiment. Such implementation-specific decisions may include, and likely are not limited to, compliance with system-related, business-related, government-related and other constraints, which may vary by specific implementation, location and from time to time. While a developer's efforts might be complex and time-consuming in an absolute sense, such efforts would be, nevertheless, a routine undertaking for those of skill in this art having benefit of this disclosure. It must be understood that the inventions disclosed and taught herein are susceptible to numerous and various modifications and alternative forms.

[0031] The use of a singular term, such as, but not limited to, "a," is not intended as limiting of the number of items. Also, the use of relational terms, such as, but not limited to, "top," "bottom," "left," "right," "upper," "lower," "down," "up," "side," and the like are used in the written description for clarity in specific reference to the figures and are not intended to limit the scope of the inventions or the appended claims. The terms "including" and "such as" are illustrative and not limitative. The terms "couple," "coupled," "coupling," "coupler," and like terms are used broadly herein and can include any method or device for securing, binding, bonding, fastening, attaching, joining, inserting therein, forming thereon or therein, communicating, or otherwise associating, for example, mechanically, magnetically, electrically, chemically, operably, directly or indirectly with intermediate elements, one or more pieces of members together and can further include without limitation integrally forming one functional member with another in a unity fashion. The coupling can occur in any direction, including rotationally. Further, all parts and components of the disclosure that are capable of being physically embodied inherently include imaginary and real characteristics regardless of whether such characteristics are expressly described herein, including but not limited to characteristics such as axes, ends, inner and outer surfaces, interior spaces, tops, bottoms, sides, boundaries, dimensions (e.g., height, length, width, thickness), mass, weight, volume and density, among others.

[0032] Any process flowcharts discussed herein illustrate the operation of possible implementations of systems, meth-

ods, and computer program products according to various embodiments of the present disclosure. In this regard, each block in a flowchart may represent a module, segment, or portion of code, which can comprise one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some implementations, the function(s) noted in the block(s) might occur out of the order depicted in the figures. For example, blocks shown in succession may, in fact, be executed substantially concurrently. It will also be noted that each block of flowchart illustration can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0033] Applicants have created new and useful devices, systems and methods for solenoid valves. By press-fitting a poppet onto a plunger, press-fitting a cap into a valve body, press-fitting a stop into a valve body, or any combination thereof, manufacturing of the valve can be simplified, parts can be eliminated, costs can be reduced, operational adjustments can be made, and/or stacked tolerances can be reduced or eliminated.

[0034] FIG. 1 is a cross sectional view of one of many embodiments of a solenoid valve according to the disclosure. FIG. 2 is a cross sectional view of a portion of one of many embodiments of a solenoid valve according to the disclosure. FIG. 3 is a cross sectional view of a portion of another one of many embodiments of a solenoid valve according to the disclosure. FIG. 4 is a cross sectional perspective view of one of many embodiments of a poppet according to the disclosure. FIG. 5 is a cross sectional view of a portion of yet another one of many embodiments of a solenoid valve according to the disclosure. FIG. 6 is a cross sectional perspective view of another one of many embodiments of a poppet according to the disclosure. FIG. 7 is a cross sectional perspective view of a portion of the poppet of FIG. 6. FIG. 8 is a cross sectional view of a portion of still another of many embodiments of a solenoid valve according to the disclosure. FIG. 9 is an exploded cross sectional view of a portion of one of many embodiments of a poppet according to the disclosure. FIG. 10 is a cross sectional view of a portion of one of many embodiments of a solenoid valve according to the disclosure. FIG. 11 is a cross sectional view of another portion of one of many embodiments of a solenoid valve according to the disclosure. FIG. 12 is a perspective view of one of many embodiments of a cap according to the disclosure. FIG. 13 is an elevation view of one of many embodiments of a stop according to the disclosure. FIG. 14 is a perspective view of the stop of FIG. 13. FIG. 15 is an elevation view of another one of many embodiments of a stop according to the disclosure. FIG. 16 is a cross sectional view illustrating one of many embodiments of an assembly method according to the disclosure, prior to the poppet being pressed onto the plunger. FIG. 17 is a cross sectional view illustrating another of many embodiments of an assembly method according to the disclosure, showing the poppet pressed onto the plunger. FIG. 18 is a cross sectional view illustrating yet another of many embodiments of an assembly method according to the disclosure, prior to the cap being pressed into the valve body. FIG. 19 is a cross sectional view illustrating still another of many embodiments of an assembly method according to the disclosure, showing the cap pressed into the valve body. FIG. 20 is a pneumatic diagram of one of many embodi-

ments of an assembly system according to the disclosure. FIGS. 1-20 are described in conjunction with one another.

[0035] In at least one embodiment, a valve 100 according to the disclosure, such as a miniature solenoid valve, can include one or more valve bodies 200, one or more plungers 300 disposed at least partially within the valve body 200, one or more poppets 400 coupled to the plunger 300, one or more end caps 500 coupled at least partially within the valve body 200, one or more stops 600 coupled at least partially within the valve body 200, or any combination thereof. In at least one embodiment, a valve 100 can include a shell 150 for housing or protecting one or more other components of the valve. In at least one embodiment, the poppet 400 can be press-fit onto the plunger 300. In at least one embodiment, the cap 500 and/or the stop 600 can limit movement of the plunger 300 and/or poppet 400. In at least one embodiment, the cap 500 and/or the stop 600 can be press-fit into the valve body 200.

[0036] In at least one embodiment, the valve body 200 can include one or more port sections 210, one or more flux collar support sections 220, one or more coil support sections 230, or any combination thereof, extending along one or more axes, such as a central longitudinal axis X. In at least one embodiment, the port section 210 can include one or more common ports 212, one or more normally open ports 214, and one or more normally closed ports 216, or any combination thereof. In at least one embodiment, the flux collar support section 220 can support a flux collar, such as to improve efficiency of the valve 100. In at least one embodiment, the port section 210, the flux collar support section 220, and the coil support section 230 of the valve body 200 can be a unitary or monolithic structure, such as by way of being portions of an integrally formed structure (e.g., an injection molded structure).

[0037] In at least one embodiment, the plunger 300 can extend at least partially through the flux collar support section 220 and the coil support section 230. In at least one embodiment, the plunger 300 can extend at least partially through a flux collar supported by the flux collar support section 220. In at least one embodiment, the plunger 300 can include one or more stems 310 that extend into the port section 210. In at least one embodiment, the plunger 300 can include one or more poppets 400 mounted to the stem 310. In at least one embodiment, one or more biasing members 330, such as a spring, can bias the plunger 300 and/or poppet 400 towards or away from one or more caps 500.

[0038] In at least one embodiment, the coil support section 230 of the valve body 200 can include one or more annular walls 232 and/or one or more coils 234. In at least one embodiment, the wall 232 can support coil 234, which can selectively actuate, or move, the plunger 300 against the force of the spring 330, to thereby switch the valve 100 from a normally open position to a normally closed position. In at least one embodiment, the coil support section 230 can include one or more stops 600, such as a core and/or related structure, to limit the movement of the plunger 300 in one or more directions within the bore 242. In at least one embodiment, the stop 600 can be a single, unitary structure. In at least one embodiment, the stop 600 can be or include a plurality of structures.

[0039] In at least one embodiment, a valve 100 according to the disclosure can include one or more poppets 400 press-fit onto the plunger 300 and configured to selectively open and close at least one flow path through the port section

210 of the valve body **200**. In at least one embodiment, the plunger **300** can include a stem **310** extending into the port section **210** of the valve body **200**. In at least one embodiment, the poppet **400** can include one or more shoulders **410** against which the spring **330** can rest and thereby bias the poppet **400** towards a normal operating position, opening a normally open flow path and/or closing a normally closed flow path through the valve body **210**. In at least one embodiment, the coil **234** can overcome the spring **330** to move the plunger **300** and poppet **400** to close a normally open flow path and/or open a normally closed flow path through the valve body **210**.

[0040] In at least one embodiment, the poppet **400** can be press-fit onto the stem **310**. In at least one embodiment, the stem **310** can include one or more annular barbs **320** configured to engage an interior of the poppet **400** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the stem **310** can include one annular barb **320** configured to engage an interior of the poppet **400** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the stem **310** can include a plurality of annular barbs **320** configured to engage an interior of the poppet **400** to hold the poppet **400** in place on the stem **310**.

[0041] In at least one embodiment, the poppet **400** can include one or more annular grooves **420** configured to engage the barbs(s) **320** on the stem **310** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the poppet **400** can include one annular groove **420** configured to engage the barbs(s) **320** on the stem **310** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the poppet **400** can include a plurality of annular grooves **420** configured to engage the barbs(s) **320** on the stem **310** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the annular barb(s) **320** and/or the annular groove(s) **420** allow a position of the poppet **400** with respect to the stem **310** to be adjusted, such that the poppet **400** can be mounted along a range of positions along the stem **310**.

[0042] In at least one embodiment, the poppet **400** can be unitary or made of multiple portions. In at least one embodiment, the poppet **400** can include a more rigid portion **430** configured to engage the barb **320** on the stem **310**. In at least one embodiment, the poppet **400** can include a more a resilient portion configured to selectively engage one or more valve seats **240**, **540** to selectively open and close the flow path through the port section **210** of the valve body **200**. In at least one embodiment, with the rigid portion **430** being harder, or more rigid, than the resilient portion **440**, the rigid portion **430** can better hold the poppet **400** in place along the stem **310** of the plunger **300**. In at least one embodiment, with the resilient portion **440** being more resilient, softer, and/or more elastomeric than the rigid portion **430**, the resilient portion **440** can better seal with the valve seats **240**, **540**. In at least one embodiment, the resilient portion **440** can be pulled at least partially into, thru, or onto the rigid portion **430**. In at least one embodiment, the rigid portion **430** can include one or more annular flanges **432**. In at least one embodiment, the resilient portion **440** can include one or more slots **442** configured to mate with the flange **432**.

[0043] In at least one embodiment, the poppet **400** can include one or more bands **450** around a perimeter of the poppet **400** to secure the poppet **400** to the stem **310**. In at least one embodiment, the poppet **400** can be pulled at least partially into, or thru, the band **450**. In at least one embodi-

ment, the band **450** can be metal and/or crimped onto the poppet **400**, thereby securing the poppet **400** to the stem **310**. In at least one embodiment, the band **450** can be a more rigid polymer than the poppet **400**, or the resilient portion **440** thereof, thereby supporting and/or securing the poppet **400** to the stem **310**.

[0044] In at least one embodiment, the poppet **400** can selectively engage one or more valve seats **240**, **540** to selectively open and close one or more flow paths through the port section **210** of the valve body **200**. In at least one embodiment, one or more valve seats **240** can be integral with the valve body **200** and/or can be part of a normally open or normally closed flow path. In at least one embodiment, one or more valve seats **540** can be integral with a cap **500** press-fit into one end of the valve body **200**.

[0045] In at least one embodiment, the poppet **400** can include one or more seals **460**, such as gasket(s) or O-ring(s), configured to selectively engage one or more valve seats **240**, **540** to selectively open and close one or more flow path through the port section **210** of the valve body **200**. In at least one embodiment, the poppet **400** can include a first seal **460** configured to selectively engage a first valve seat **540** to selectively open and close a normally closed flow path through the port section **210** of the valve body **200**. In at least one embodiment, the poppet **400** can include a second seal **460** configured to selectively engage a second valve seat **240** to selectively open and close a normally open flow path through the port section **210** of the valve body **200**. In at least one embodiment, the seal(s) **460** can be mounted to the poppet **400** and/or the stem **310**, with the poppet **400** selectively pressing the seals(s) **460** into the valve seats **240**, **540**.

[0046] In at least one embodiment, portions of the valve body **200** and/or the cap **500** can be resilient. In at least one embodiment, all, or portions, of the poppet **400** can be rigid. In at least one embodiment, the poppet **400** can be rigid and configured to selectively engage a resilient valve seat **240**, **540** to selectively open and close the flow path through the port section **210** of the valve body **200**.

[0047] In at least one embodiment, a valve **100** according to the disclosure, such as a miniature solenoid valve, can include one or more valve bodies **200** having a port section **210** and a coil support section **230** extending along a longitudinal axis, one or more plungers **300** within the valve body **200** and extending at least partially through the coil support section **230**, a poppet **400** including a resilient portion **440** configured to selectively engage a rigid valve seat **240**, **540** to open and close at least one flow path through the port section **210** of the valve body **200**, or any combination thereof. In at least one embodiment, the plunger **300** can include one or more stems **310** extending into the port section **210** of the valve body **200**. In at least one embodiment, the stem **310** can include at least one annular barb **320** configured to engage an interior of the poppet **400** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, one or more of the valve seats **240** can be integral with the valve body **200**. In at least one embodiment, one or more of the valve seats **540** can be integral with a cap **500** press-fit into one end of the valve body **200**.

[0048] In at least one embodiment, the poppet 400 can include one or more rigid portions 430 configured to engage the barb 320 on the stem 310. In at least one embodiment, the rigid portion 430 can be harder than the resilient portion 440. In at least one embodiment, the rigid portion 430 can include one or more annular flanges 432. In at least one embodiment, the resilient portion 440 can be configured to mate with the flange 432 and/or can include one or more slots 442 configured to mate with the flange 432.

[0049] In at least one embodiment, a valve 100 according to the disclosure, such as a miniature solenoid valve, can include one or more valve bodies 200 having a port section 210 and a coil support section 230 extending along a longitudinal axis, one or more plungers 300 within the valve body 200 and extending at least partially through the coil support section 230, one or more poppets 400 configured to selectively engage one or more valve seats 240, 540 to open and close one or more flow path through the port section 210 of the valve body 200, or any combination thereof. In at least one embodiment, the plunger 300 can include a stem 310 extending into the port section 210 of the valve body 200. In at least one embodiment, the stem 310 can include one or more annular bars 320. In at least one embodiment, the bars 320 can engage an interior of the poppet 400 to hold the poppet 400 in place on the stem 310.

[0050] By press-fitting the poppet 400 onto the stem 310 of the plunger 300, manufacturing of the valve 100 can be simplified, parts can be eliminated, costs can be reduced, operational adjustments can be made, and/or stacked tolerances can be reduced or eliminated. For example, a poppet body or holder can be eliminated. In at least one embodiment, the poppet 400 according to the disclosure does not require a complex and/or expensive over molding process. In at least one embodiment, the poppet 400 according to the disclosure can include a myriad of techniques, as described above, to provide adequate holding force, or resistance to pull-out, to remain in place on the plunger 300 for proper functionality within the valve 100. In at least one embodiment, the poppet 400 according to the disclosure can be a monolithic or unitary elastomeric, or otherwise resilient, structure with integral shoulder 410 for the spring 330 to provide a bias force against in order to operate the valve 100. In at least one embodiment, one or more of the bars 320 and/or grooves 420 can be spiral, thereby allowing the poppet 400 according to the disclosure to be press-fit and/or threaded onto the stem 310, while still achieving many or all of the advantages discussed herein.

[0051] In at least one embodiment, an external surface of the stem 310 and/or an internal surface of the poppet 400 can be rough, or roughened, to increase static friction between the stem 310 and the poppet 400. In at least one embodiment, a dissolvable and/or evaporable lubricant can be used during assembly of the valve 100, such as to temporarily reduce friction between the stem 310 and the poppet 400. In at least one embodiment, an adhesive can be used to bond the stem 310 and the poppet 400. In at least one embodiment, the poppet 400 can include a through hole in order to inject an adhesive between the stem 310 and the poppet 400. In at least one embodiment, the poppet 400 can be designed to allow for transfer of heat or ultraviolet light to the adhesive between the stem 310 and the poppet 400. In at least one embodiment, the poppet 400 according to the disclosure can be vulcanized, hardened, shrunk, or any combination thereof once in place on the stem 310. In at

least one embodiment, the poppet 400 according to the disclosure, or a portion thereof, can be metal and/or a rigid polymer and/or the valve seats 240, 540 can be metal and/or a rigid polymer, with the valve 100 relying on a metal-to-metal and/or metal-to-plastic seal, thereby eliminating elastomeric materials in sealing the flow path(s).

[0052] In at least one embodiment, a valve 100 according to the disclosure can include one or more caps 500 press-fit at least partially into the valve body 200 at an end of the valve body 200, such as at the port section 210 of the valve body. In at least one embodiment, the cap 500 can include one or more valve seats 540 that can be selectively engaged by the poppet 400 mounted to the plunger 300 to selectively open and close a flow path through the port section 210 of the valve body 200. In at least one embodiment, the cap 500 can include one or more annular bars 520 that sealingly mate with the valve body 200. In at least one embodiment, the barb 520 can extend around a perimeter of the cap 500. In at least one embodiment, the barb 520 can resist pressure within the valve body 200 to hold the cap 500 in place within or relative to the valve body 500. In at least one embodiment, the annular barb(s) 520 can allow a position of the cap 500 within the valve body 200 to be adjusted, such that the cap 500 can be mounted along a range of positions along the valve body 200. Adjusting the position of the cap 500 within or relative to the valve body 200 can adjust the stroke of the plunger 300, one or more flow rates through the port section 210 of the valve body, reduce stacked tolerances in the valve 100, or any combination thereof.

[0053] In at least one embodiment, the barb 520 can be integral with the cap 500 and/or can hermetically or fluidically seal the cap 500 to the valve body 200, which can include doing so without any additional sealing device, such as an O-ring, gasket, or sealant. In at least one embodiment, the barb 520 and/or the cap 500 can be metal and/or the valve body 200 can be a resilient material. In at least one embodiment, the barb 520 can deform the valve body 200, such as while being press-fit therein, thereby sealing the cap 500 to the valve body 200.

[0054] In at least one embodiment, the cap 500 can include one annular barb 520 that extends around a perimeter of the cap 500. In at least one embodiment, the cap 500 can include two or more annular bars 520 that extend around a perimeter of the cap 500. In at least one embodiment, the bars 520 can resist pressure within the valve body 200 to hold the cap 500 in place within or relative to the valve body 200.

[0055] In at least one embodiment, a valve 100 according to the disclosure can include one or more stops 600 press-fit at least partially into the valve body 200 at an end of the valve body 200, such as at the coil section 230 of the valve body 200. In at least one embodiment, the stop 600 can include one or more shafts 610 to limit movement of the plunger 300 away from (i.e., in a direction away from) the cap 500 and/or a flange 612, which can fix a position of the stop 600 with respect to the valve body 200. In at least one embodiment, the stop 600 can include one or more annular bars 620 that sealingly mates with the valve body 200. In at least one embodiment, the barb 620 can extend around a perimeter of the stop 600. In at least one embodiment, the barb 620 can extend around a perimeter of the shaft 610 and/or the flange 612. In at least one embodiment, the barb 620 can resist pressure within the valve body 200 to hold the stop 600 in place within or relative to the valve body 200. In at least one embodiment, the annular barb(s) 620 can

allow a position of the stop 600 within or relative to the valve body 200 to be adjusted, such that the stop 600 can be mounted along a range of positions along the valve body 200. Adjusting the position of the stop 600 within or relative to the valve body 200 can adjust the stroke of the plunger 300, and/or reduce stacked tolerances in the valve 100.

[0056] In at least one embodiment, the barb 620 can be integral with the stop 600 and/or can hermetically or fluidically seal the stop 600 to the valve body 200, which can include doing so singlehandedly, such as without the aid of an additional sealing device, such as an O-ring, gasket, or sealant. In at least one embodiment, the barb 620 and/or the stop 600 can be metal and/or the valve body 200 can be a resilient material. In at least one embodiment, the barb 620 can be configured to deform the valve body 200, such as while being press-fit therein, thereby sealing the stop 600 to the valve body 200.

[0057] In at least one embodiment, the stop 600 can include one or more annular barbs 620 that extend around a perimeter of the stop 600. In at least one embodiment, the stop 600 can include two or more annular barbs 620 that extend around a perimeter of the stop 600. In at least one embodiment, the barbs 620 can resist pressure within the valve body 200 and/or can hold the stop 600 in place within or relative to the valve body 200.

[0058] In at least one embodiment, a valve 100 according to the disclosure can include one or more valve bodies 200 having a port section 210 and a coil support section 230 extending along a longitudinal axis, a plunger 300 within the valve body 200 and extending at least partially through the coil support section 230, a cap 500 press-fit at least partially into the valve body 200 at a first end of the valve body 200, a stop 600 press-fit at least partially into the valve body 200 at a second, opposite end of the valve body 200, or any combination thereof. In at least one embodiment, the cap 500 can include a valve seat 540 configured to be selectively engaged by a poppet 400 mounted to the plunger 300 to selectively open and close a flow path through the port section 210 of the valve body 200. In at least one embodiment, the cap 500 can include one or more first annular barbs 520 extending around a perimeter of the cap 500, which can sealingly secure the cap 500 at least partially within the valve body 200. In at least one embodiment, the stop 600 can limit movement of the plunger 300 away from the cap 500. In at least one embodiment, the stop 600 can include one or more second or other annular barbs 620 extending around a perimeter of the stop 600, which can sealingly secure the stop 600 at least partially within the valve body 200.

[0059] In at least one embodiment, the first barb 520 can resist pressure within the valve body 200 to hold the cap 500 in place within or relative to the valve body 200. In at least one embodiment, the first barb 520 can be metal. In at least one embodiment, the first barb 520 and the cap 520 can be integrally formed of metal. In at least one embodiment, the valve body 200 can be a resilient material. In at least one embodiment, the first barb 520 can deform the valve body 200, such as while being press-fit therein, thereby sealing the cap 500 to the valve body 200.

[0060] In at least one embodiment, the second barb 620 can resist pressure within the valve body 200 to hold the stop 600 in place within or relative to the valve body 200. In at least one embodiment, the barb can be metal. In at least one embodiment, the second barb and the stop can be integrally formed of metal. In at least one embodiment, the valve body

can be a resilient material. In at least one embodiment, the second barb can deform the valve body, such as while being press-fit therein, thereby sealing the stop 600 to the valve body.

[0061] Utilizing barbs 520, 620 on the cap 500 and/or stop 600, any of which can be metallic, advantageously provides retention and/or hermetic sealing within the valve body 200, which can be plastic or polymer, that requires no additional parts to function, reduces the number of parts required, reduces time and/or effort required for assembly, allows for positional fixation and/or adjustment within the valve body 200, or any combination thereof. In at least one embodiment, the cap 500 can include two or more barbs 520 to prevent angular misalignment of the valve seat 540 to the valve body 200 and/or plunger 400. In at least one embodiment, heat can be used to shrink, stake, and/or further couple or seal the valve body 200 to the cap 500 and/or the stop 600.

[0062] In at least one embodiment, a valve 100 according to the disclosure can include one or more valve bodies 200 having a port section 210 and a coil support section 230 extending along a longitudinal axis, a plunger 300 within the valve body 200 and extending at least partially through the coil support section 230, a poppet 400 press-fit onto the plunger 300, or any combination thereof. In at least one embodiment, the body 200 can include one or more first valve seats 240. In at least one embodiment, the plunger 300 can include a stem 310 extending through the first valve 240 seat into the port section 210. In at least one embodiment, the poppet 400 can be press-fit onto the stem 310 of the plunger 400. In at least one embodiment, the poppet 400 can selectively engage the first valve seat 240 to open and close a first flow path through the port section 210 of the valve body 200. In at least one embodiment, a stroke of the plunger 300 and/or a flow rate through the first flow path can be adjusted by adjusting a first position of the poppet 400 on the plunger 300.

[0063] In at least one embodiment, a valve 100 according to the disclosure can include one or more caps 500 press-fit at least partially into the port section 210 of the valve body 200. In at least one embodiment, the cap 500 can include one or more second valve seat 540. In at least one embodiment, the poppet 400 can selectively engage the second valve seat 540 to open and close a second flow path through the port section 210 of the valve body 200. In at least one embodiment, a stroke of the plunger 300 and/or a flow rate through the second flow path can be adjusted by adjusting a second position of the cap 500 in or relative to the body 200.

[0064] In at least one embodiment, a method of assembling a valve 100 according to the disclosure can include monitoring a first flow, of fluid flow, between a first port and a second port of the valve 100 and/or pressing a poppet 400 onto a plunger 300 of the valve 100 until the first flow stops. In at least one embodiment, a method of assembling a valve 100 according to the disclosure can include monitoring a second fluid flow between the first port and a third port of the valve 100, energizing a coil 234 of the valve 100, pressing a cap 500 into a body 200 of the valve 100 until the second fluid flow stops, or any combination thereof. In at least one embodiment, a method of assembling a valve 100 according to the disclosure can include monitoring a second fluid flow between the first port and a third port of the valve 100 and/or pressing a cap 500 into a body 200 of the valve 100 until a target rate of the second fluid flow is achieved, which can be or include any flow rate required or desired according to an

implementation of the disclosure. In at least one embodiment, the first port can be a common port. In at least one embodiment, the second port can be a normally closed port. In at least one embodiment, the third port can be a normally open port.

[0065] In at least one embodiment, an assembly system or apparatus **700**, such as that shown in FIG. **20** for exemplary purposes, can be used to provide a controlled fluid flow to/from one or more ports of the port section **210** of the valve body **200** during assembly of valve **100**. In at least one embodiment, such fluid flow can be of air, another inert gas, water, another inert fluid, or any combination thereof. In at least one embodiment, the system **700** can be or include a pneumatic system, one or more assembly fixtures (not shown) and one or more presses or pressing machines (not shown), such as a dynamic precision press capable of precision pressing of components based on closed loop sensor feedback. In at least one embodiment, the system **700** can one or more supply pressure inlets, one or more solenoid valves (e.g., solenoid valves SV1, SV2, et seq.), one or more pressure regulators (e.g., pressure regulators PR1, PR2), one or more flow meters (e.g., flow meter FM), one or more level meters (e.g., level meter LM), one or more pressure transmitters (e.g., pressure transmitter PT), one or more orifices, one or more valve port connections, one or more vents to atmosphere, or any combination thereof.

[0066] In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include monitoring a power consumption of the coil **234** and continuing to press the cap **500** into the body **200** and/or the poppet **400** onto the plunger **300**, until the power consumption matches a setpoint, such as a setpoint indicative of the assembly step being complete. By adjusting the position of the cap **500** in or relative to the body **200** and/or the poppet **400** on the plunger **300**, power consumption of the coil **234** can be controlled.

[0067] In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include energizing a coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, and monitoring for a sound caused by a plunger **300** of the valve **100** contacting a stop **600** of the valve **100**. In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include repeating the steps of pressing the poppet **400** onto the plunger **300**, energizing the coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, monitoring for the sound caused by the plunger **300** contacting the stop **600**, or any combination thereof, until the sound caused by the plunger **300** contacting the stop **600** is no longer detected. By adjusting the position of the poppet **400** on the plunger **300**, the stroke of the plunger **300** can be controlled, such as to limit the engagement between the plunger **300** and the stop **600**.

[0068] In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include monitoring a first fluid flow between two or more ports, such as a common port **212** and a normally closed port **216** of the valve **100**, pressing a poppet **400** onto a plunger **300** of the valve **100** until the first fluid flow stops, monitoring a second fluid flow between the common port **212** and a normally open port **214** of the valve **100**, energizing a coil **234** of the valve **100**, pressing a cap **500** into a body **200** of the valve **100** until the second fluid flow stops, or any combination thereof. In at least one embodiment, a method of assembling

a valve **100** according to the disclosure can include monitoring a power consumption of the coil **234** and/or continuing to press, or repeatedly pressing, the cap **500** into the body **200** and/or the poppet **400** onto the plunger **300**, until the power consumption matches a setpoint, which can be or include any setpoint required or desired according to an implementation of the disclosure. In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include energizing the coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, monitoring for a sound caused by a plunger **300** of the valve **100** contacting one or more other components, such as a stop **600** of the valve **100**, or any combination thereof. In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include repeatedly pressing the poppet **400** onto the plunger **300**, energizing the coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, monitoring for the sound caused by the plunger **300**, or any combination thereof, until the sound caused by the plunger **300** is no longer detected, falls below a threshold, or falls within an acceptable range, which can be or include any threshold or range required or desired according to an implementation of the disclosure. In at least one embodiment, the poppet **400** can be pressed onto the plunger **300** before the cap **500** is pressed into the body **200**.

[0069] In at least one embodiment, a method of assembling a valve **100** can include monitoring a fluid flow between the common port and a normally open port of the valve **100**, energizing a coil **234** of the valve **100**, pressing a cap **500** into a body **200** of the valve **100** until the fluid flow stops, or any combination thereof. In at least one embodiment, the method can include monitoring a power consumption of the coil **234** and repeating the step of pressing the cap **500** into the body **200**, until the power consumption matches a setpoint. In at least one embodiment, the method can include energizing a coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, monitoring for a sound caused by a plunger **300** of the valve **100** contacting a stop **600** of the valve **100**, or any combination thereof. In at least one embodiment, the method can include repeating, until the sound caused by the plunger **300** contacting the stop **600** is no longer detected, the steps of pressing a poppet **400** onto the plunger **300**, energizing the coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, monitoring for the sound caused by the plunger **300** contacting the stop **600**, or any combination thereof. In at least one embodiment, the poppet **400** can be pressed onto the plunger **300** before the cap **500** is pressed into the body **200**.

[0070] By monitoring flow rates through the port section **210** of the valve **100** while pressing the poppet **400** onto the plunger **300** and/or the cap **500** into the body **200** the position of the poppet **400** on the plunger **300** and/or the cap **500** in the body **200** can be adjusted, thereby reducing, eliminating, or otherwise mitigating the effects of stack-up tolerances in the valve **100**, such as those impacting the stroke of the plunger **300**. Pressing the poppet **400** further onto the plunger **300** and/or pressing the cap **500** further into the body **200** can adjust the stroke of the plunger **300**, flow through the port section of the valve **100**, power consumption of the coil **234**, operational noise within the valve **100**, load and/or preload on the spring **330**, or any combination thereof. The assembly method described herein can reduce costs through increased tolerances (i.e., the adjustability

reduces the effect of tolerances, thereby allowing for looser tolerances in individual component manufacturing) provide better performance by reducing excess elastomer deformation on the valve seats **240**, **540**, mitigate against energized leak by conducting a pressing operation until enough elastomer deformation has occurred to adequately seal between the cap **500** and the body **200** and/or between the stop **600** and the body **200**, offer precise flow control, offer variability in flow rates without change to orifice size, increase reliability of the valve **100** (e.g., since functionality can be confirmed during assembly), or any combination thereof. Stack-up tolerances can be mitigated through the use of the annular barbs **320**, **520**, **620** described herein, threaded barbs, parts sorting (such as to counter tolerances), or any combination thereof.

[0071] In at least one embodiment, a valve can include a valve body having a port section and a coil support section extending along a longitudinal axis, a plunger within the valve body and extending at least partially through the coil support section, a poppet press-fit onto the plunger, or any combination thereof. In at least one embodiment, the body can include a first valve seat. In at least one embodiment, the plunger can include a stem extending through the first valve seat into the port section. In at least one embodiment, the poppet can selectively engage the first valve seat to open and close a first flow path through the port section of the valve body. In at least one embodiment, a stroke of the plunger can be adjusted by adjusting a position of the poppet on the plunger. In at least one embodiment, a flow rate through the first flow path is configured to be adjusted by adjusting a position of the poppet on the plunger.

[0072] In at least one embodiment, the valve can include a cap press-fit at least partially into the port section of the valve body. In at least one embodiment, the cap can include a second valve seat that is configured to be selectively engaged by the poppet to selectively open and close a second flow path through the port section of the valve body. In at least one embodiment, the stroke of the plunger can be adjusted by adjusting a second position of the cap in the body. In at least one embodiment, a flow rate through the second flow path can be adjusted by adjusting a second position of the cap in the body.

[0073] In at least one embodiment, a method of assembling a valve can include monitoring a first flow between a first port and a second port of the valve and/or pressing a poppet onto a plunger of the valve until the first flow stops. In at least one embodiment, the method can include monitoring a second fluid flow between the first port and a third port of the valve, energizing a coil of the valve, pressing a cap into a body of the valve until the second fluid flow stops, or any combination thereof. In at least one embodiment, the method can include monitoring a second fluid flow between the first port and a third port of the valve and/or pressing a cap into a body of the valve until a target rate of the second fluid flow is achieved. In at least one embodiment, the first port is a common port, the second port is a normally closed port, and the third port is a normally open port.

[0074] In at least one embodiment, the method can include monitoring a power consumption of the coil and repeating the step of pressing the cap into the body, until the power consumption matches a setpoint. In at least one embodiment, the method can include energizing a coil of the valve, de-energizing the coil of the valve, monitoring for a sound caused by a plunger of the valve contacting a stop of the

valve, or any combination thereof. In at least one embodiment, the method can include repeating, until the sound caused by the plunger contacting the stop is no longer detected, the steps of pressing the poppet onto the plunger, energizing the coil of the valve, de-energizing the coil of the valve, monitoring for the sound caused by the plunger contacting the stop, or any combination thereof.

[0075] In at least one embodiment, a method of assembling a valve can include monitoring a first flow between a common port and a normally closed port of the valve, pressing a poppet onto a plunger of the valve until the first flow stops, monitoring a second fluid flow between the common port and a normally open port of the valve, energizing a coil of the valve, pressing a cap into a body of the valve until the second fluid flow stops, or any combination thereof. In at least one embodiment, the method can include monitoring a power consumption of the coil and repeating the step of pressing the cap into the body, until the power consumption matches a setpoint. In at least one embodiment, the method can include energizing a coil of the valve, de-energizing the coil of the valve, monitoring for a sound caused by a plunger of the valve contacting a stop of the valve, or any combination thereof. In at least one embodiment, the method can include repeating, until the sound caused by the plunger contacting the stop is no longer detected, the steps of pressing the poppet onto the plunger, energizing the coil of the valve, de-energizing the coil of the valve, monitoring for the sound caused by the plunger contacting the stop, or any combination thereof. In at least one embodiment, the poppet can be pressed onto the plunger before the cap is pressed into the body.

[0076] In at least one embodiment, a method of assembling a valve can include monitoring a fluid flow between two or more ports of the valve, energizing a coil of the valve, pressing a cap into a body of the valve until the fluid flow stops, or any combination thereof. In at least one embodiment, the method can include monitoring a power consumption of the coil and repeating the step of pressing the cap into the body, until the power consumption matches a setpoint. In at least one embodiment, the method can include energizing a coil of the valve, de-energizing the coil of the valve, monitoring for a sound caused by a plunger of the valve contacting one or more other components, such as a stop of the valve, or any combination thereof. In at least one embodiment, the method can include repeating, until a sound caused by the plunger is no longer detected (or, e.g., falls below a threshold or within an acceptable range), the steps of pressing a poppet onto a plunger of the valve, energizing the coil of the valve, de-energizing the coil of the valve, monitoring for the sound caused by the plunger, or any combination thereof. In at least one embodiment, the poppet can be pressed onto the plunger before the cap is pressed into the body.

[0077] Other and further embodiments utilizing one or more aspects of the disclosure can be devised without departing from the spirit of Applicants' disclosure. For example, the devices, systems and methods can be implemented for numerous different types and sizes in numerous different industries. Further, the various methods and embodiments of the devices, systems and methods can be included in combination with each other to produce variations of the disclosed methods and embodiments. Discussion of singular elements can include plural elements and vice versa. The order of steps can occur in a variety of sequences

unless otherwise specifically limited. The various steps described herein can be combined with other steps, interlineated with the stated steps, and/or split into multiple steps. Similarly, elements have been described functionally and can be embodied as separate components or can be combined into components having multiple functions.

[0078] The inventions have been described in the context of preferred and other embodiments and not every embodiment of the inventions has been described. Obvious modifications and alterations to the described embodiments are available to those of ordinary skill in the art having the benefits of the present disclosure. The disclosed and undisclosed embodiments are not intended to limit or restrict the scope or applicability of the inventions conceived of by the Applicants, but rather, in conformity with the patent laws, Applicants intend to fully protect all such modifications and improvements that come within the scope or range of equivalents of the following claims.

1. A method of assembling a valve, the method comprising:

monitoring a first fluid flow between a first port and a second port of the valve; and
pressing a poppet onto a plunger of the valve until the first fluid flow stops.

2. The method of claim 1, wherein the first port is a common port and the second port is a normally closed port.

3. The method of claim 1, further comprising
monitoring a second fluid flow between the first port and a third port of the valve;
energizing a coil of the valve; and
pressing a cap into a body of the valve until the second fluid flow stops.

4. The method of claim 3, wherein the first port is a common port, the second port is a normally closed port, and the third port is a normally open port.

5. The method of claim 3, further comprising
monitoring a power consumption of the coil; and
repeating, until the power consumption matches a set-point, the step of pressing the cap into the body of the valve.

6. The method of claim 1, further comprising
monitoring a second fluid flow between the first port and a third port of the valve; and
pressing a cap into a body of the valve until a target rate of the second fluid flow is achieved.

7. The method of claim 6, wherein the first port is a common port, the second port is a normally closed port, and the third port is a normally open port.

8. The method of claim 6, further comprising
monitoring a power consumption of the coil; and
repeating, until the power consumption matches a set-point, the step of pressing the cap into the body.

9. The method of claim 1, further comprising
energizing a coil of the valve;
de-energizing the coil of the valve; and
monitoring for a sound caused by a plunger of the valve contacting a stop of the valve.

10. The method of claim 9, further comprising
repeating, until the sound caused by the plunger contacting the stop is no longer detected, the steps of
pressing the poppet onto the plunger;
energizing the coil of the valve;
de-energizing the coil of the valve; and

monitoring for the sound caused by the plunger contacting the stop.

11. A method of assembling a valve, the method comprising:

monitoring a first fluid flow between a common port and a normally closed port of the valve;
pressing a poppet onto a plunger of the valve until the first fluid flow stops;
monitoring a second fluid flow between the common port and a normally open port of the valve;
energizing a coil of the valve; and
pressing a cap into a body of the valve until the second fluid flow stops.

12. The method of claim 11, further comprising
monitoring a power consumption of the coil; and
repeating, until the power consumption matches a set-point, the step of pressing the cap into the body.

13. The method of claim 11, further comprising
energizing the coil of the valve;
de-energizing the coil of the valve; and
monitoring for a sound caused by a plunger of the valve contacting a stop of the valve.

14. The method of claim 13, further comprising
repeating, until the sound caused by the plunger contacting the stop is no longer detected, the steps of
pressing the poppet onto the plunger;
energizing the coil of the valve;
de-energizing the coil of the valve; and
monitoring for the sound caused by the plunger contacting the stop.

15. The method of claim 14, wherein the steps of claim 14 are performed before the step of pressing the cap into the body.

16. A method of assembling a valve, the method comprising:

monitoring a fluid flow between a common port and a normally open port of the valve;
energizing a coil of the valve; and
pressing a cap into a body of the valve until the fluid flow stops.

17. The method of claim 16, further including
monitoring a power consumption of the coil; and
repeating, until the power consumption matches a set-point, the step of pressing the cap into the body.

18. The method of claim 16, further including
energizing the coil of the valve;
de-energizing the coil of the valve; and
monitoring for a sound caused by a plunger of the valve contacting a stop of the valve.

19. The method of claim 18, further including
repeating, until the sound caused by the plunger contacting the stop is no longer detected, the steps of
pressing a poppet onto a plunger of the valve;
energizing the coil of the valve;
de-energizing the coil of the valve; and
monitoring for the sound caused by the plunger contacting the stop.

20. The method of claim 19, wherein the steps of claim 19 are performed before the step of pressing the cap into the body.